LSE Data Analytics Career Accelerator Technical Report

Background

This project was an exercise in using predictive analytics and machine learning modelling to better understand patterns and trends within the Turtle Reviews data.

I focused on loyalty points and how other customer metrics related to and with such points.

I wanted to address six main areas of interest in my analysis:

Q1) Possible linear relationships between age, remuneration, and spending score regarding loyalty point accumulation?

Q2) How well can decision trees be used to understand loyalty point accumulation?

Q3) Can k-means clustering aid in understanding trends within remuneration and spending score?

Q4) Can multiple linear regression tell us anything about the relationship between remuneration, spending score and loyalty points?

Q5) How is a logistic regression model useful in showing how likely customers are to be 'high' loyalty point individuals?

Q6) How can natural language-processing be used to understand customer sentiments?

Methodology

1. **Data source:** 'turtle_reviews.csv'.
2. **Data selection and analysis:** The file was uploaded to Jupyter Notebook (via VSC) and RStudio., for analysis.
    a. Further transformation of the source data was used to produce 'reviews_clean.csv', keeping only relevant variables:

```
reviews.drop('language', axis=1, inplace=True)
reviews.drop('platform', axis=1, inplace=True)
reviews.rename(columns={'remuneration (k£)': 'rem'}, inplace=True)
reviews.rename(columns={'spending_score (1-100)': 'spending_score'}, inplace=True)
```

3. **Data presentation/visualisation:** Data presentation was conducted using libraries/packages such as matplotlib, seaborn and ggplot2.

I have explained my methodology in detail my working and thought processes within my .ipynb and .R files. My explanations were explained via markdown and in-line comments.

Q1. Possible linear relationships between age, remuneration, and spending score regarding loyalty point accumulation?

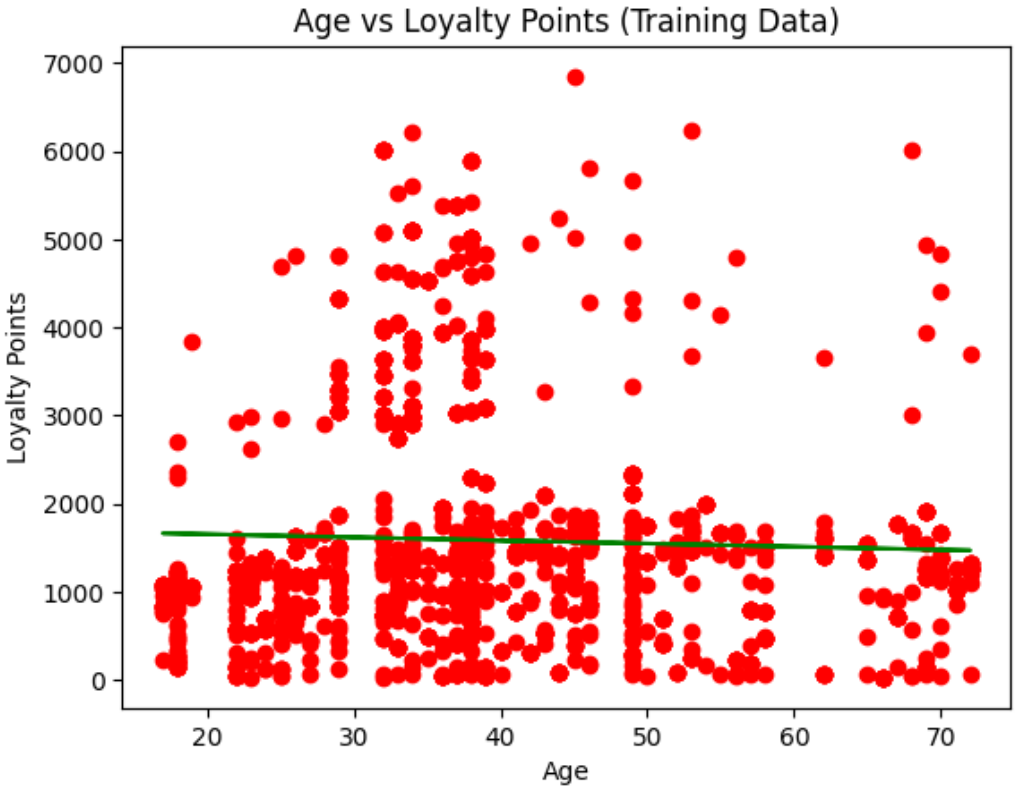
I created three separate simple linear regression models, whereby:

$$y = \beta 0 + \beta 1x + \epsilon$$

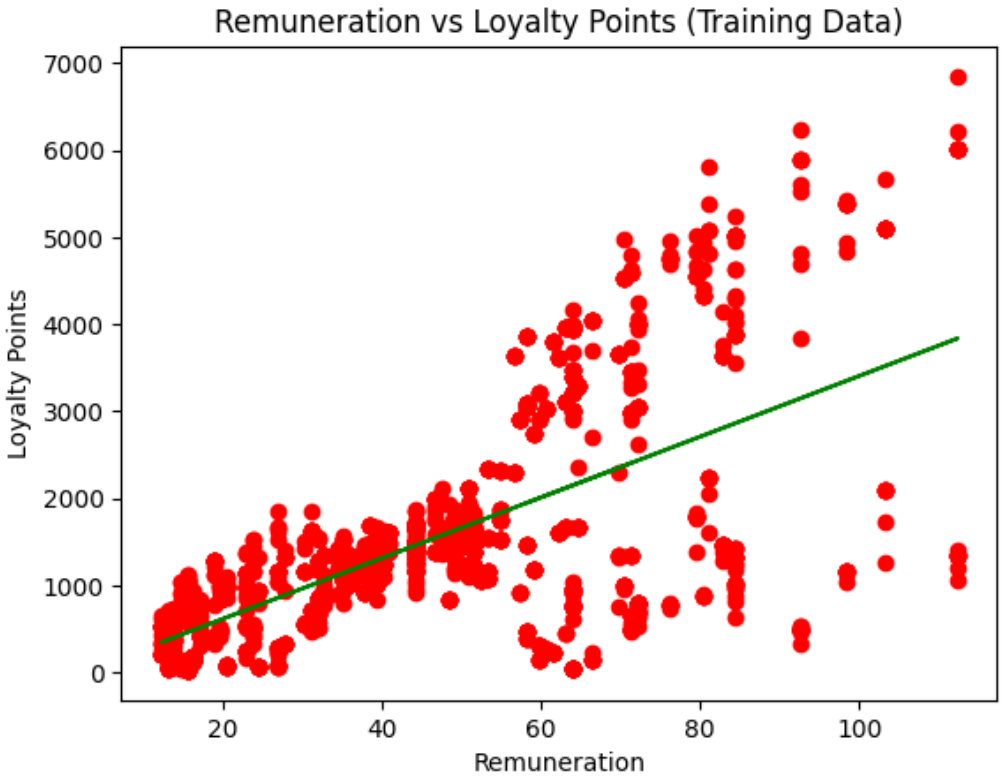
- Model 1:
  $y = loyaltypoints, x = age$
- Model 2:
  $y = loyaltypoints, x = remuneration$
- Model 3:
  $y = loyaltypoints, x = spendingscore$

<u>Initial model creation</u>

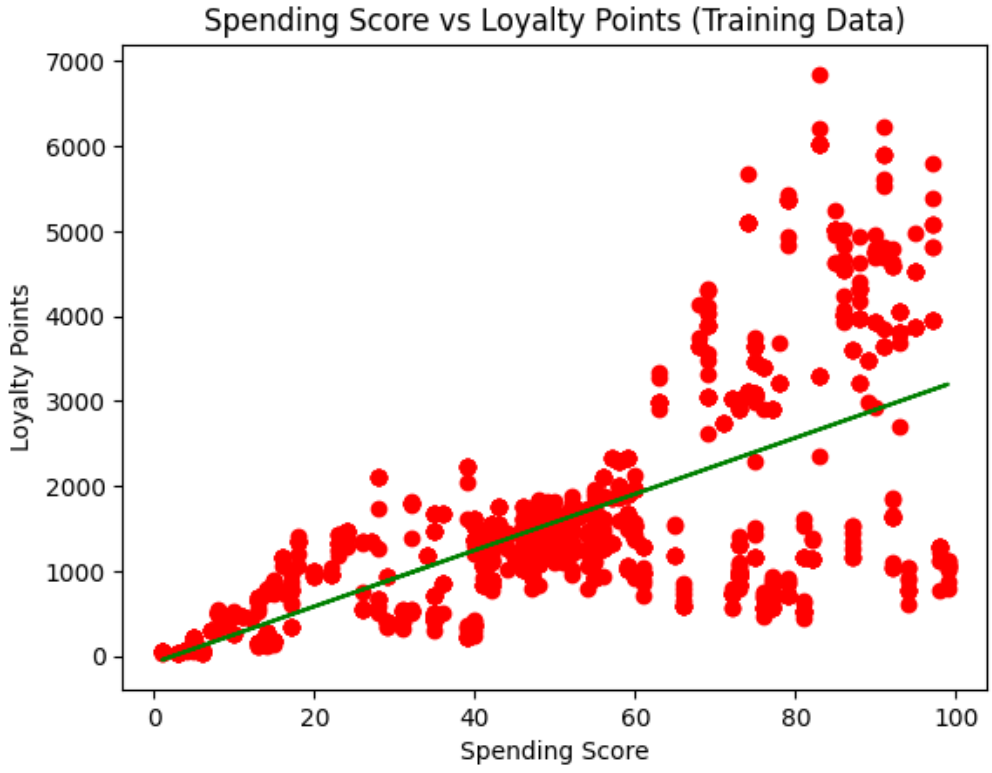
For each model, I made an initial split (80:20) and produced a scatterplot on the training data with a line of best fit to explore the relation between X and Y.



n=1600


Remuneration vs Loyalty Points (Training Data)
Loyalty Points
Remuneration
0
1000
2000
3000
4000
5000
6000
7000
20
40
60
80
100

n=1600


Spending Score vs Loyalty Points (Training Data)
Loyalty Points
Spending Score
0
1000
2000
3000
4000
5000
6000
7000
0
20
40
60
80
100

n=1600

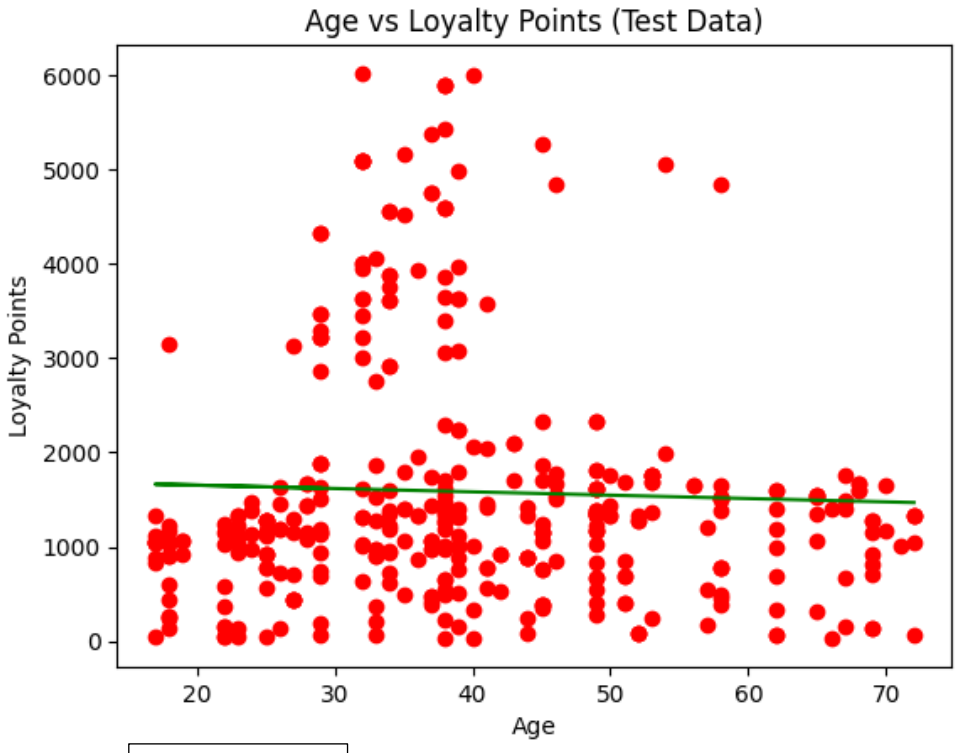
Model analysis (training set):

Model	R-Squared	Coefficient Value
Model 1	0.001	-3.545
Model 2	0.394	34.899
Model 3	0.448	33.114

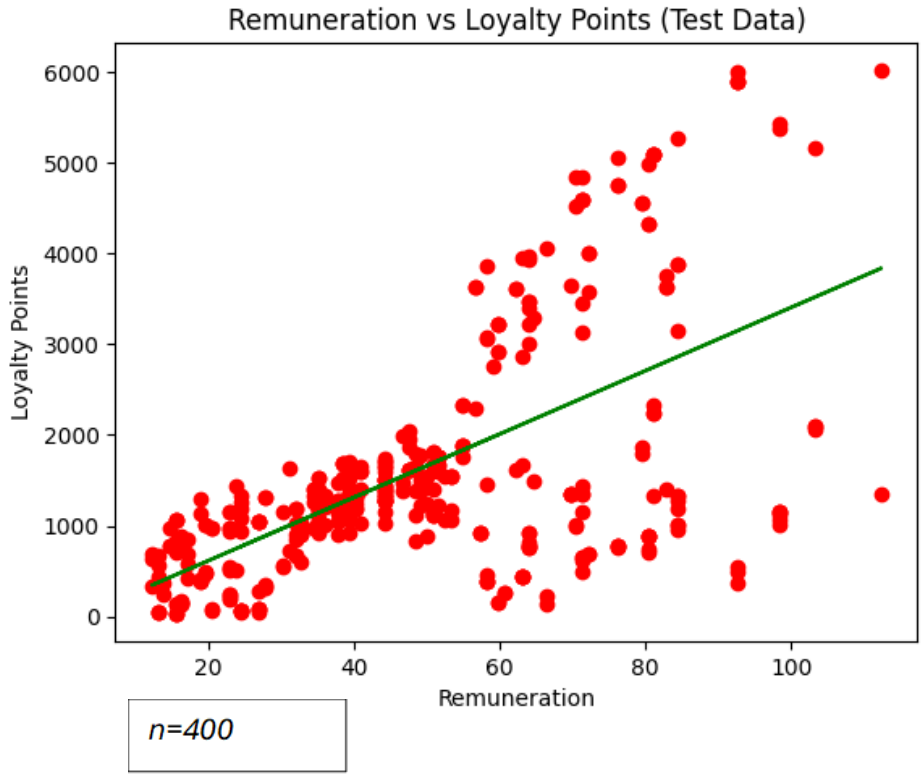
- Model 1 has a very low R-Squared. This indicates age explains little of the variability in loyalty points. A linear regression model does not fit the data well.
- Model 2 R-Squared value suggests a moderate fit. Remuneration captures some variability in loyalty points, but not all. The coefficient value indicates a positive linear relationship.
- Model 3 R-Squared value suggests a significantly better, yet still moderate fit. Spending score has a stronger positive correlation to loyalty point accumulation than either age or remuneration.

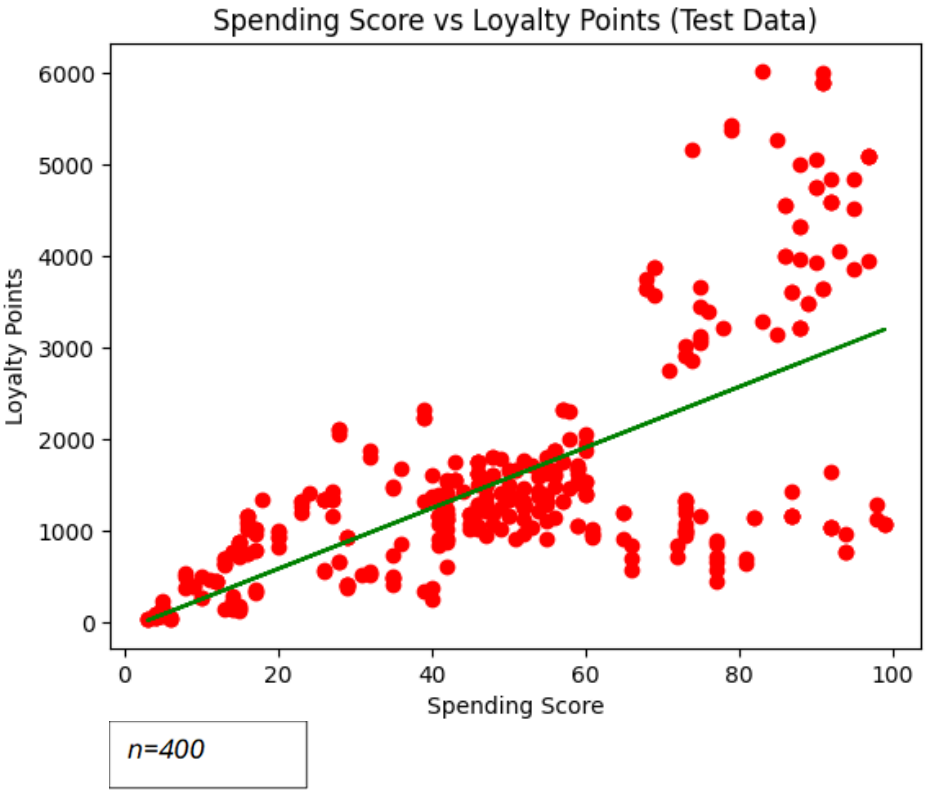
Evaluating the models on unseen test data

Following training my models on the training set, I tested their predictive ability on the test set.



n=400


Remuneration vs Loyalty Points (Test Data)
Loyalty Points
Remuneration
0
1000
2000
3000
4000
5000
6000
20
40
60
80
100
n=400


Spending Score vs Loyalty Points (Test Data)
Loyalty Points
Spending Score
0
1000
2000
3000
4000
5000
6000
0
20
40
60
80
100
n=400

Model analysis (test set)

Model	R-Squared	MAE	M.SE	RMSE
Model 1	0.003	905.481	1617180.521	1271.684
Model 2	0.318	748.464	1106063.543	1051.696
Model 3	0.466	651.696	865341.581	930.237

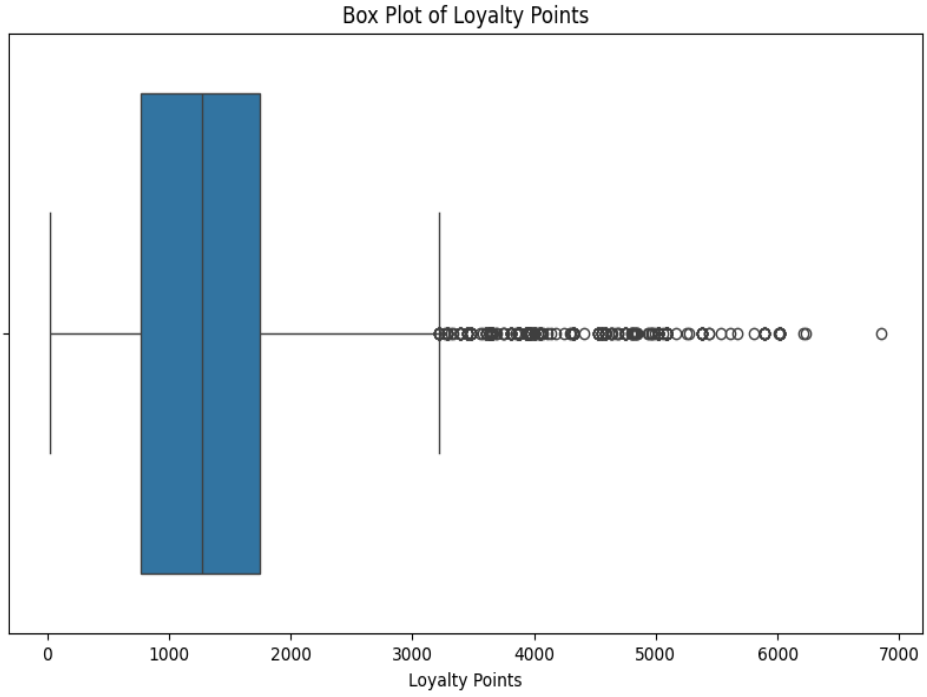
- Model 1 has a very low R-Squared, and very low scores on prediction metrics, indicating significant differences between predicted and actual values. Model 1 performs poorly and does not capture any real trends/patterns between X and Y.
- Model 2 has an R-Squared which indicates a moderate fit on unseen data, however, prediction metrics indicate significant prediction error.
- Model 3 performs better than Models 1 and 2, however, there is still discrepancy between predicted and actual Y values.

My deduction from this was that to build improved simple linear regression models for each X, data wrangling/transformation was necessary.

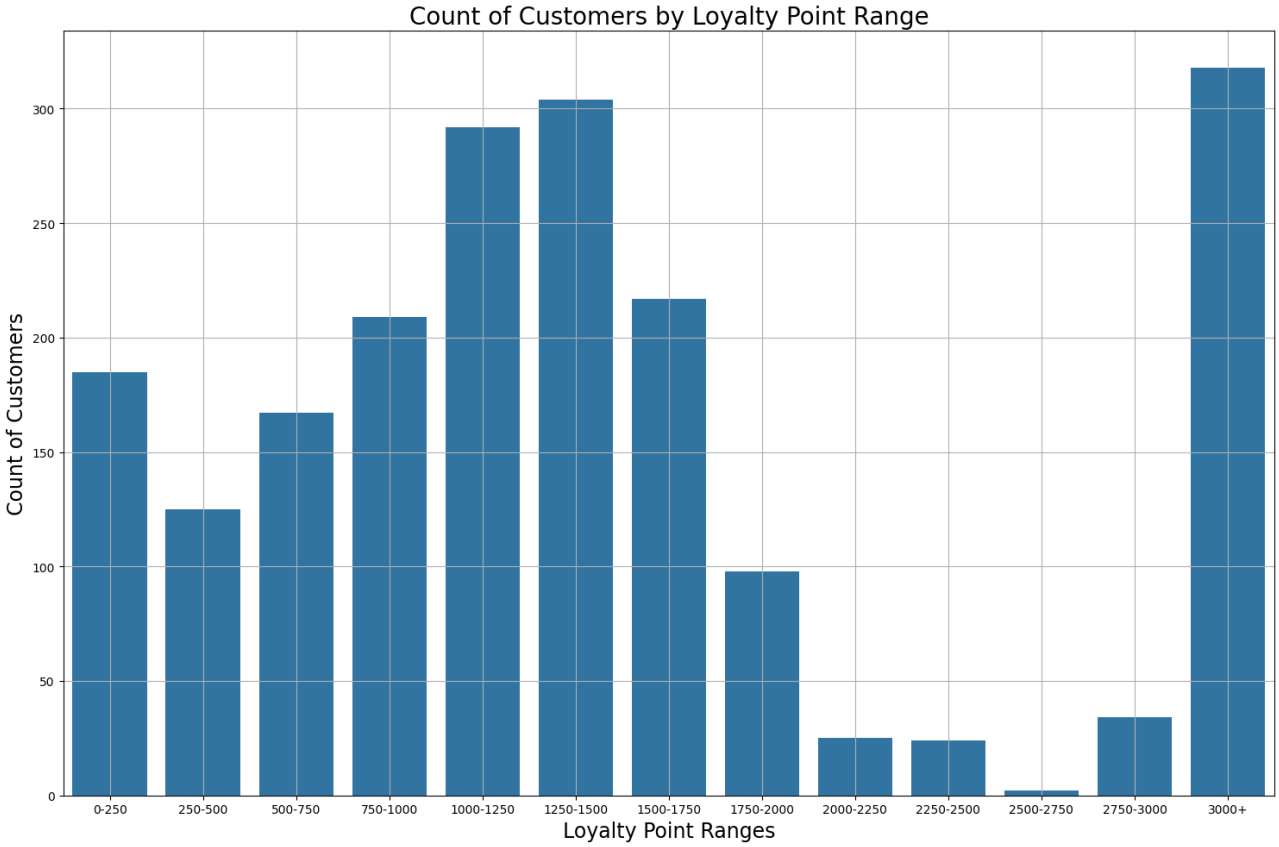
Exploring the variables

Loyalty point distribution

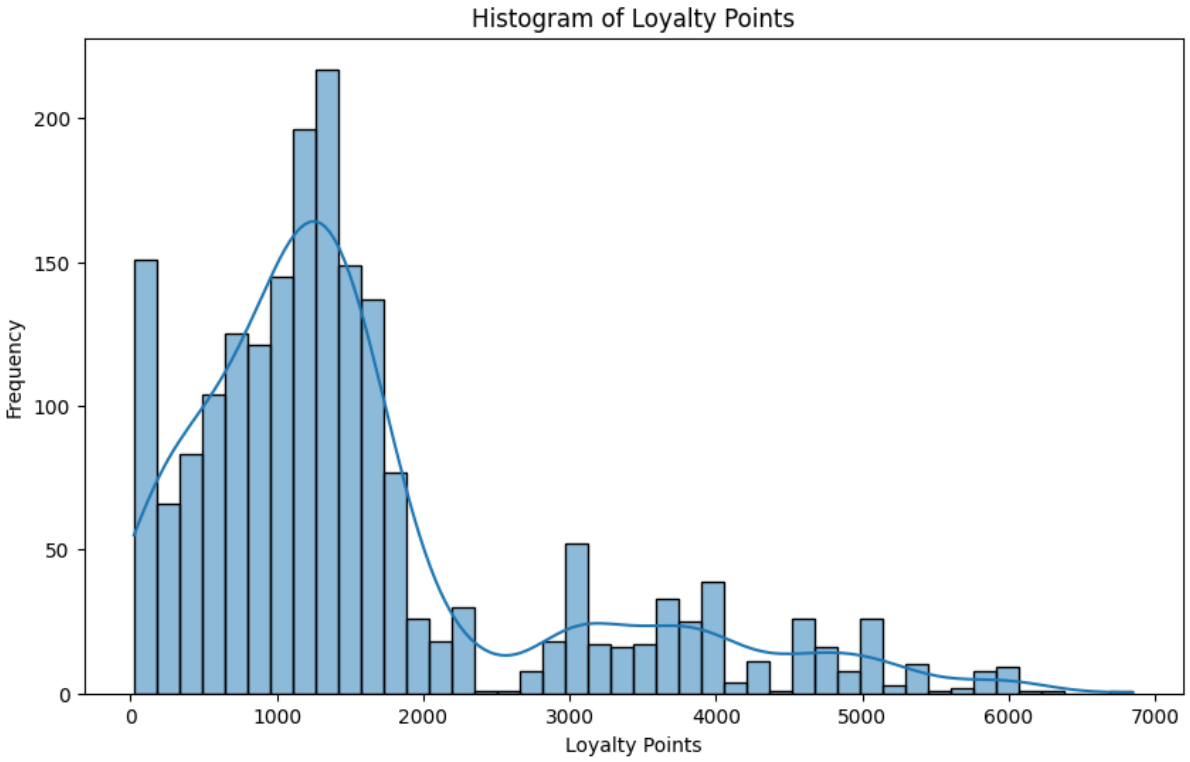
The first step in achieving this was to explore the data of the variables used.



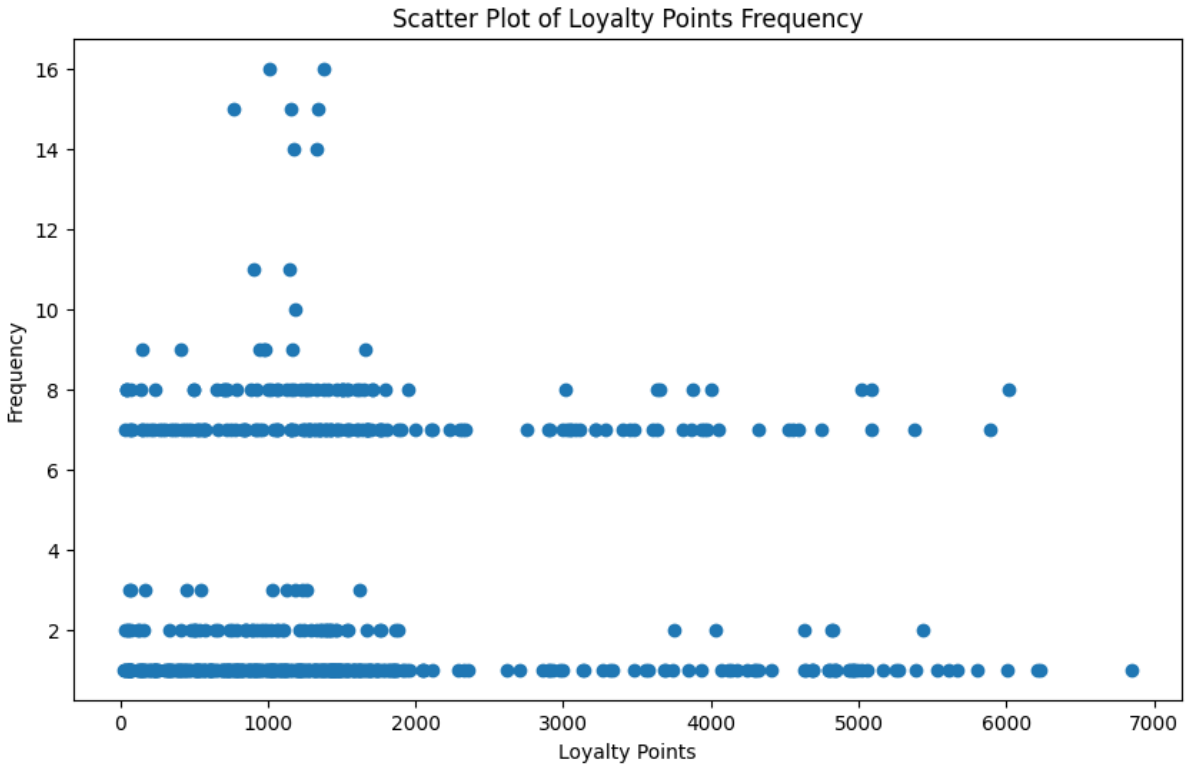
n=2000


Count of Customers by Loyalty Point Range
Count of Customers
0
50
100
150
200
250
300
0-250
250-500
500-750
750-1000
1000-1250
1250-1500
1500-1750
1750-2000
2000-2250
2250-2500
2500-2750
2750-3000
3000+
Loyalty Point Ranges

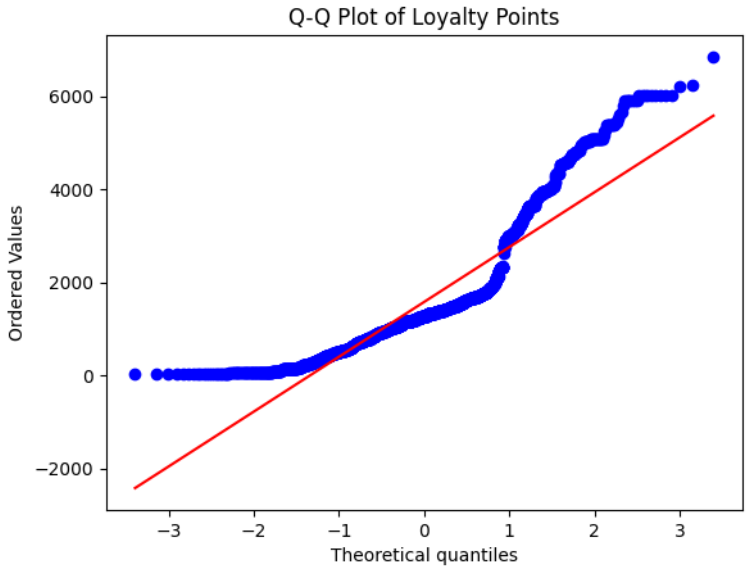
n=2000


Histogram of Loyalty Points
Frequency
0
50
100
150
200
0
1000
2000
3000
4000
5000
6000
7000
Loyalty Points

n=2000



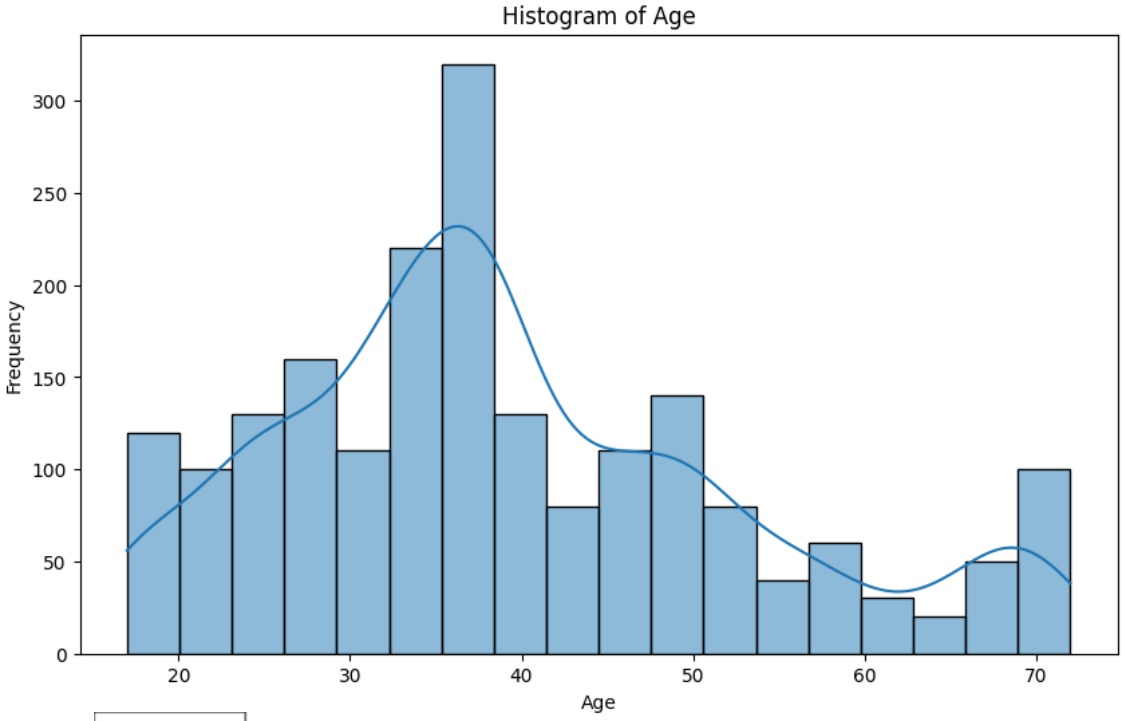
n=2000



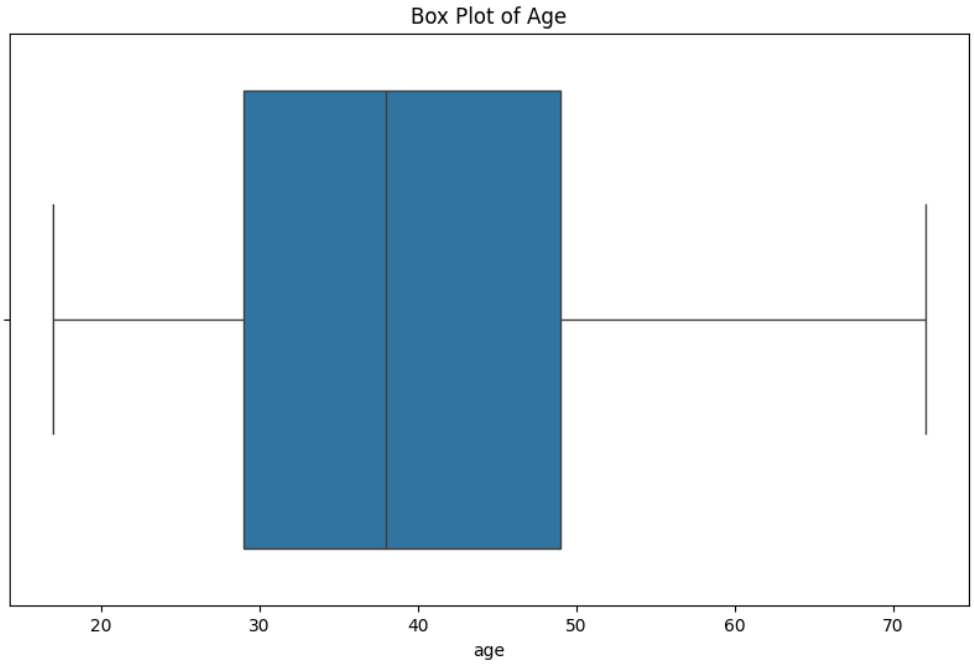
- We see greatly varied and unequal distribution of loyalty point values within the dataset.
- Most customers have loyalty points in the 0-2000 range, however, a significant contingent have above 3000, with relatively few in the 2000-3000 range. It may be suggested that **two separate populations** of loyalty point earners are present.
- The Q-Q Plot right tail indicates extreme outliers. Excess kurtosis produces such heavy right tails. However, the left tail suggests a normal distribution.

I employed a Shapiro-Wilk Test, which showed a *p-value*=1.243e-40, which also concludes loyalty point values are abnormally distributed.

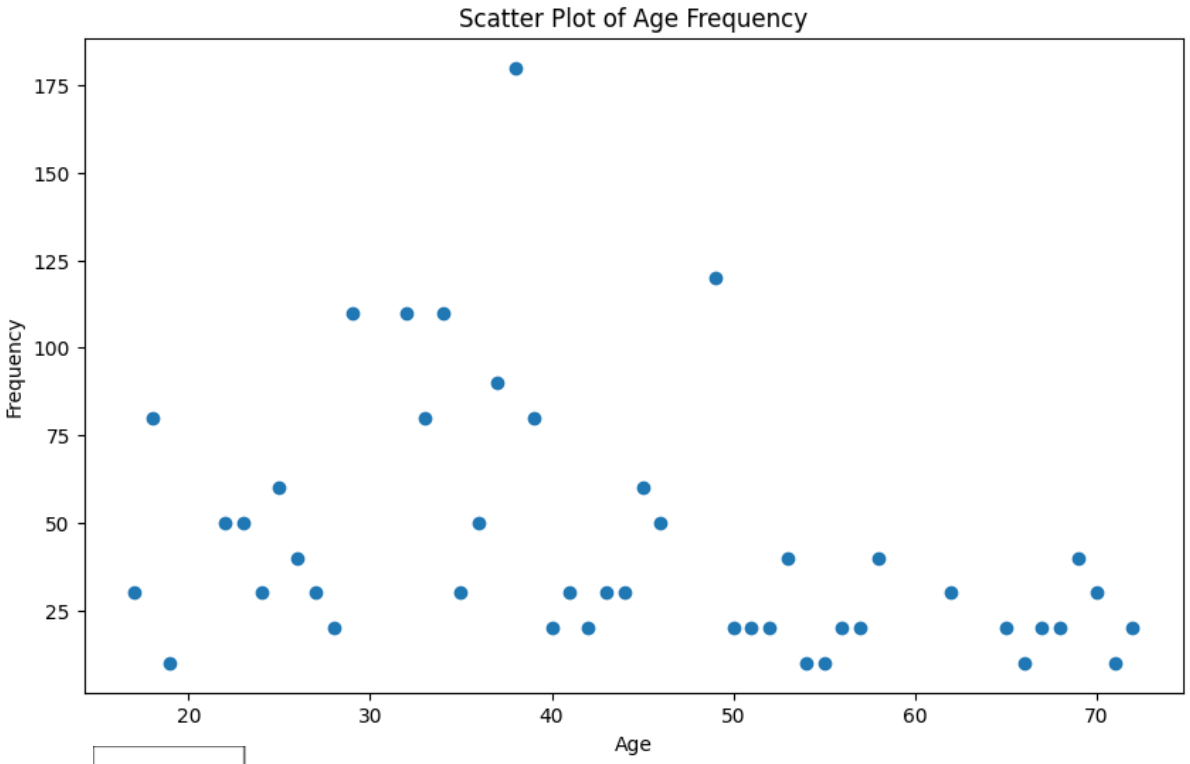
Age distribution



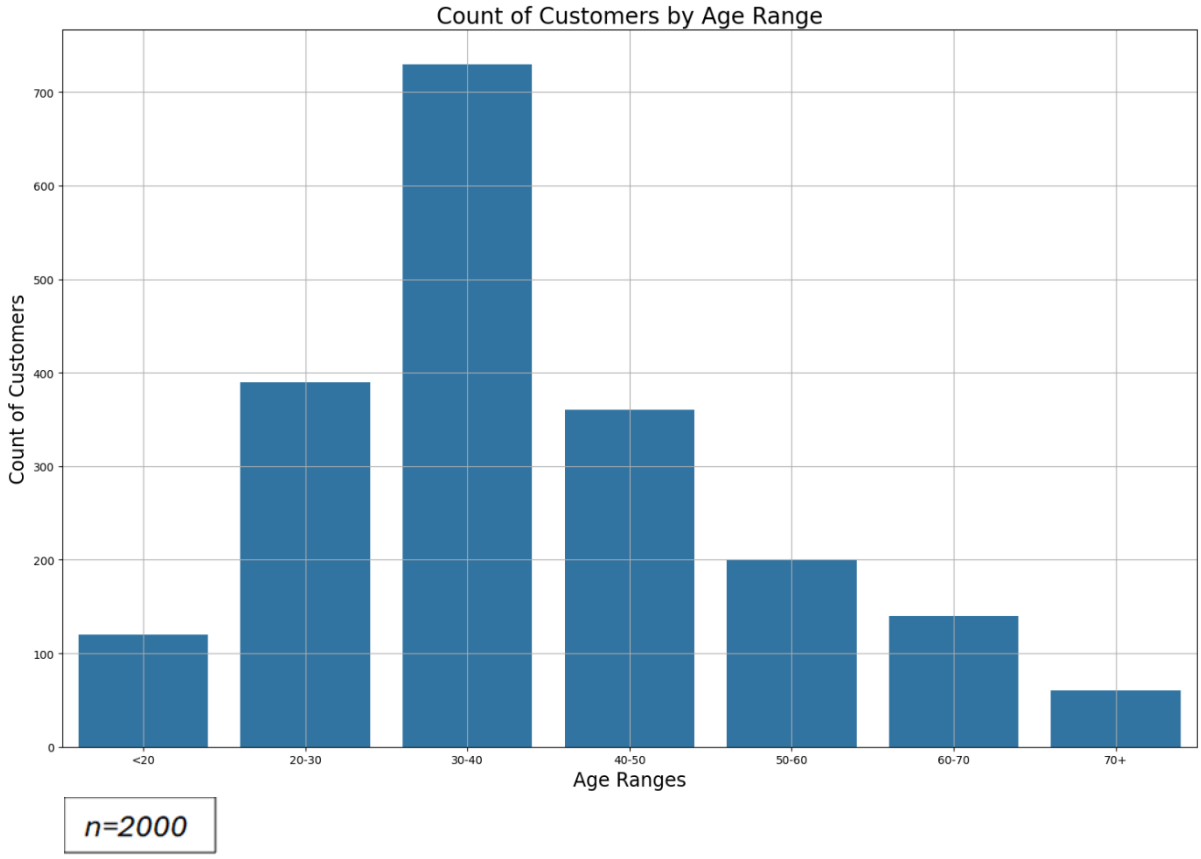
n=2000



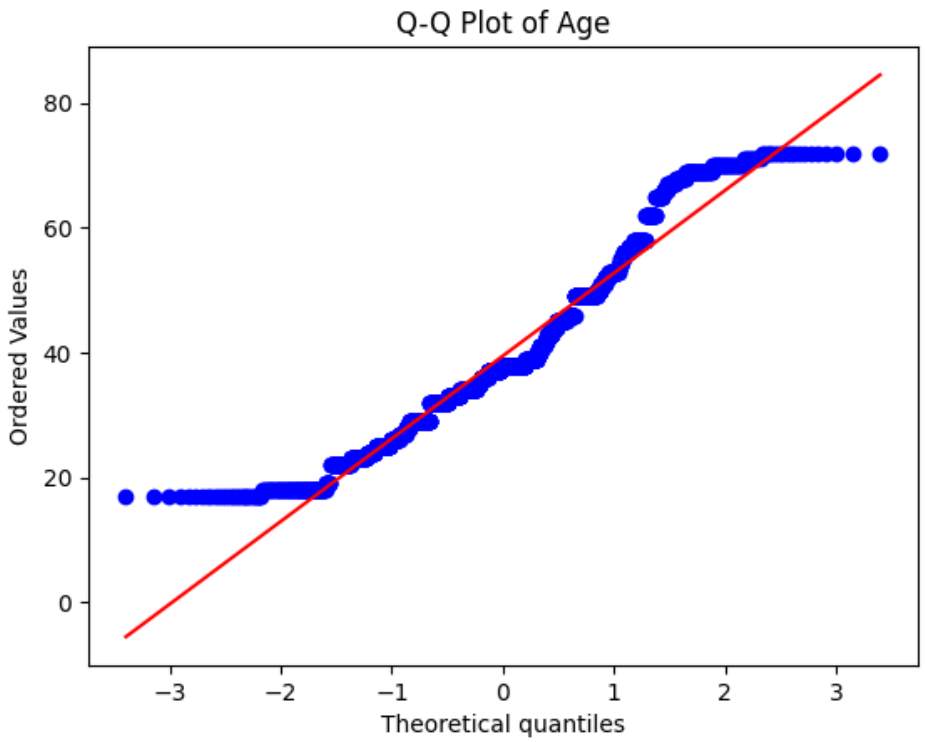
n=2000


Scatter Plot of Age Frequency
Frequency
Age
175
150
125
100
75
50
25
20
30
40
50
60
70

n=2000


Count of Customers by Age Range
Count of Customers
Age Ranges
700
600
500
400
300
200
100
0
<20
20-30
30-40
40-50
50-60
60-70
70+

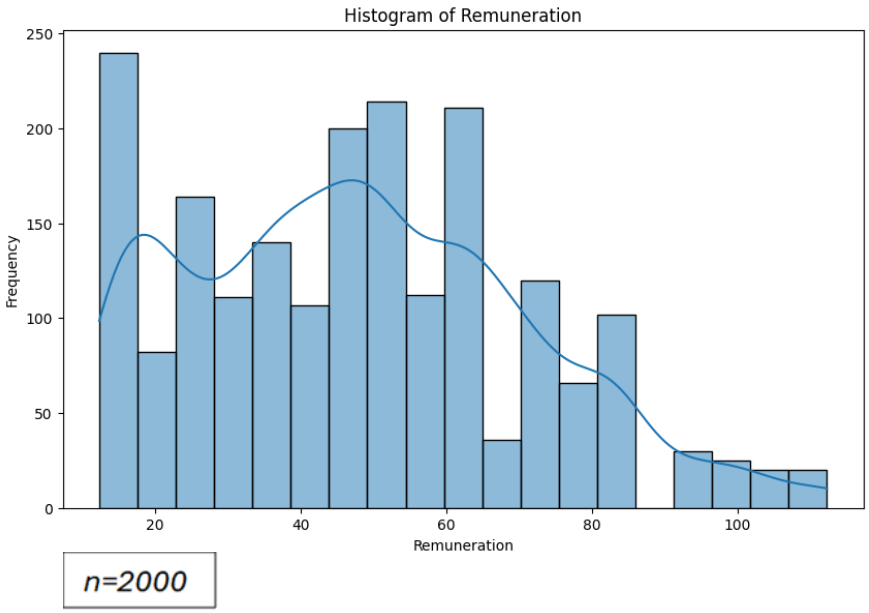
n=2000

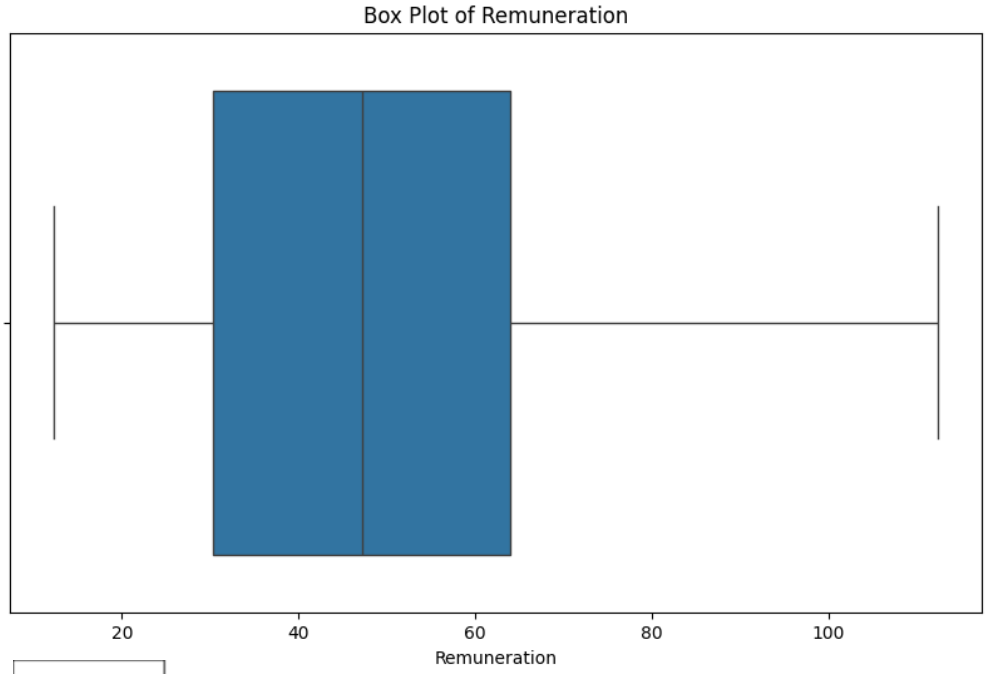


- Age has a more normal distribution than loyalty points, showing a slight negative skew.
- However, heavy right tails indicate a significant number of elderly customers (outliers).

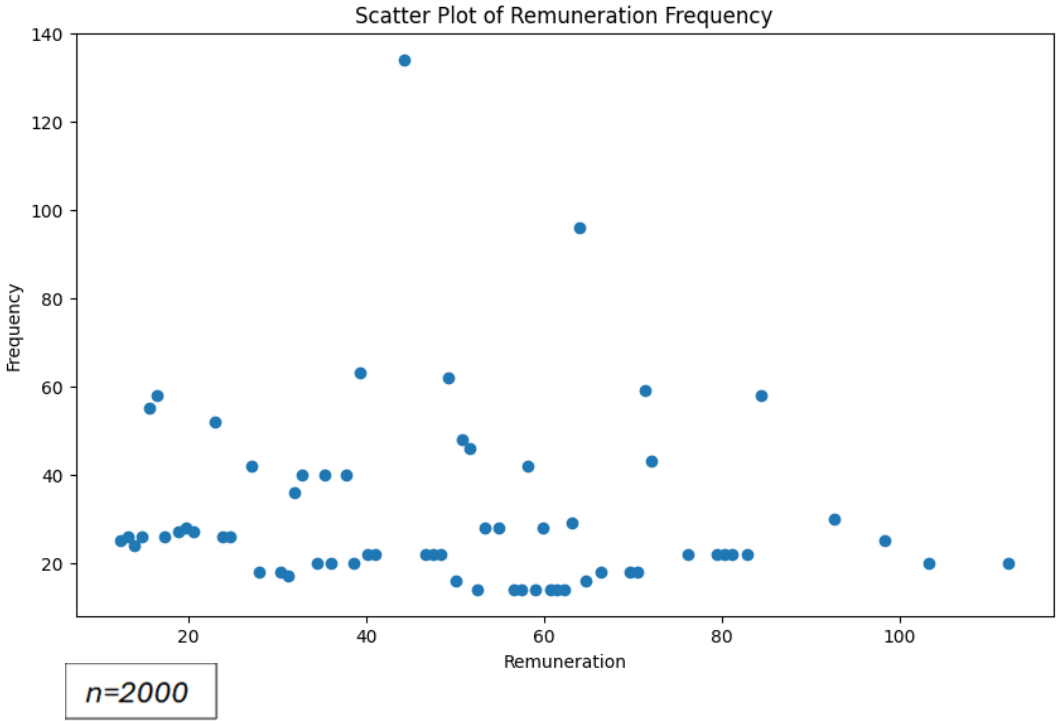
Shapiro-Wilk Test *p-value*=5.394e-25 indicated a greater normality of distribution than loyalty points, while still abnormal, however.

Remuneration distribution

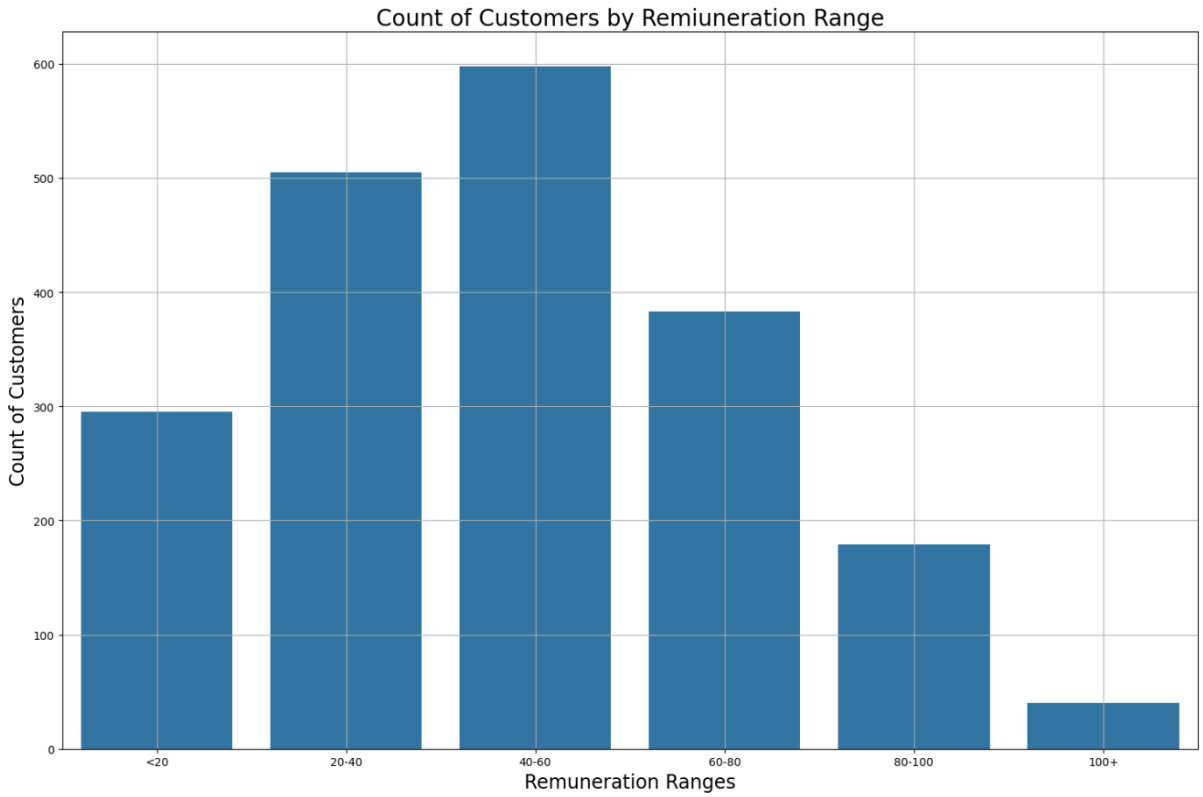



Box Plot of Remuneration
20
40
60
80
100
Remuneration

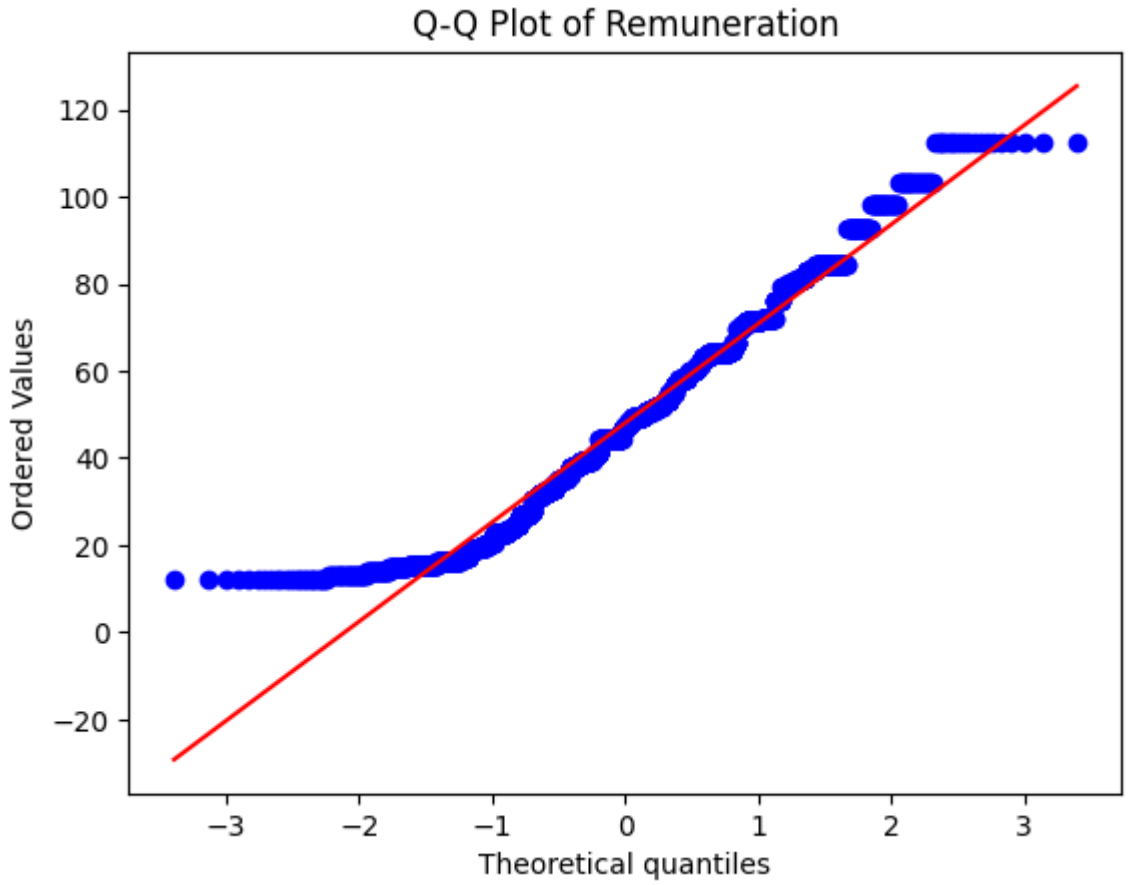
n=2000


Scatter Plot of Remuneration Frequency
Frequency
140
120
100
80
60
40
20
20
40
60
80
100
Remuneration

n=2000


Count of Customers by Remiuneration Range
Count of Customers
Remuneration Ranges
<20
20-40
40-60
60-80
80-100
100+
0
100
200
300
400
500
600

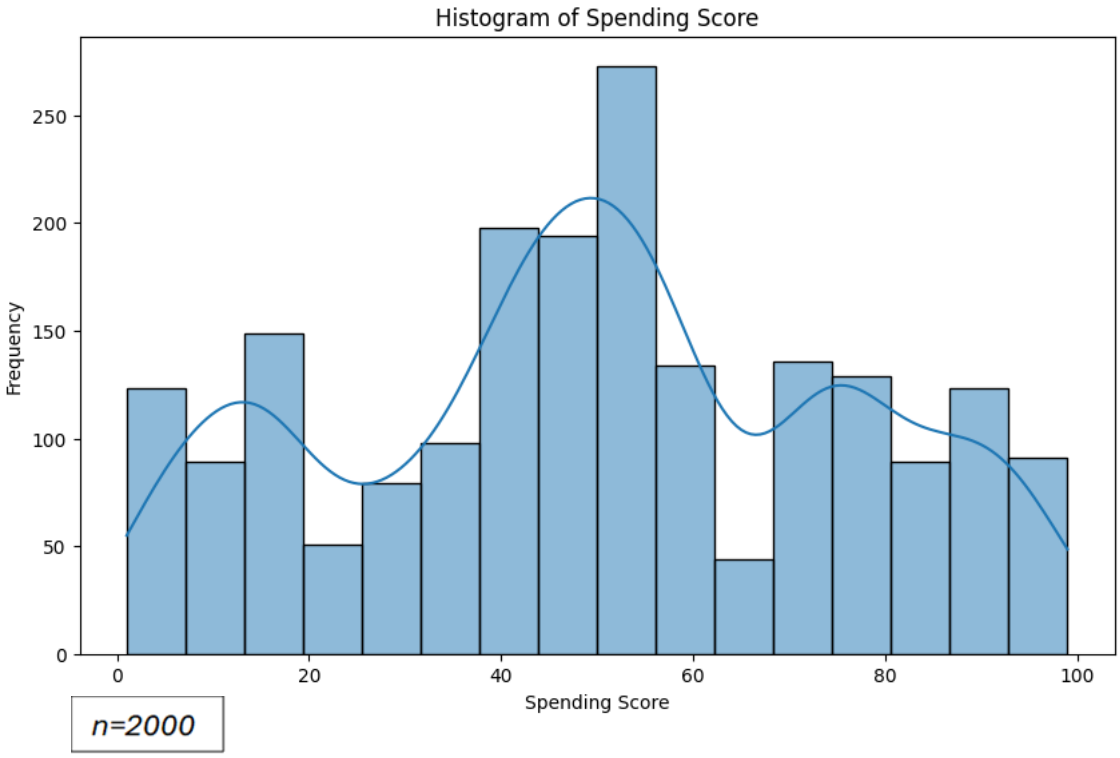
n=2000

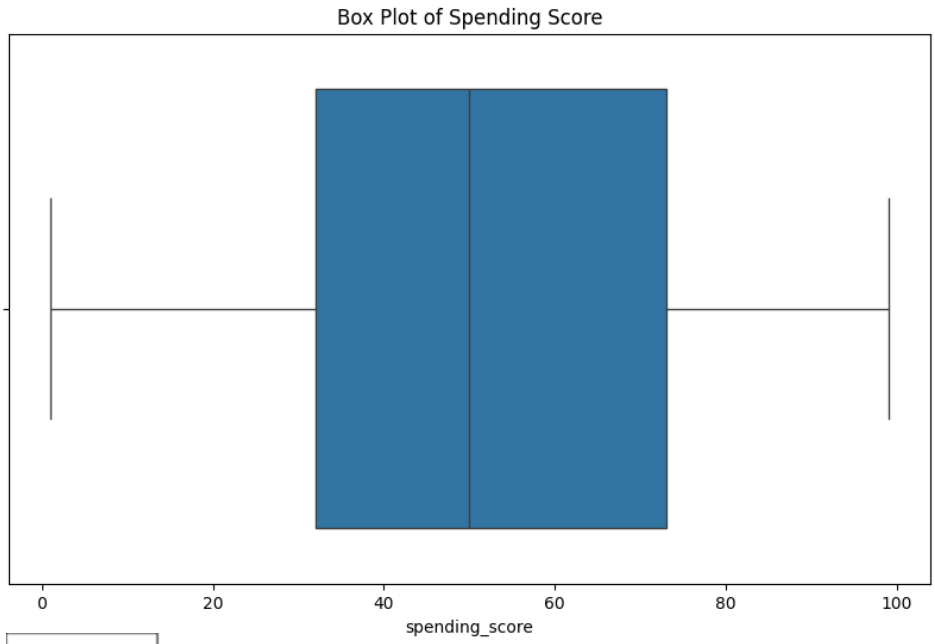

Q-Q Plot of Remuneration
Ordered Values
Theoretical quantiles
120
100
80
60
40
20
0
−20
−3
−2
−1
0
1
2
3

- Remuneration appears to be relatively normally distributed with no extreme outliers. The most popular range of customer remuneration is £40k-£60k.
- Distribution shows a slight negative skew.
- Q-Q Plot has a left tail lighter than normal distribution, indicating negative skew. Upper tail suggests the presence of outliers (>£100k). Boxplot corroborates this.

Shapiro-Wilk Test *p-value*=8.837e-21 indicates abnormal distribution of data, though not as severe as age or loyalty points.

Spending Score distribution

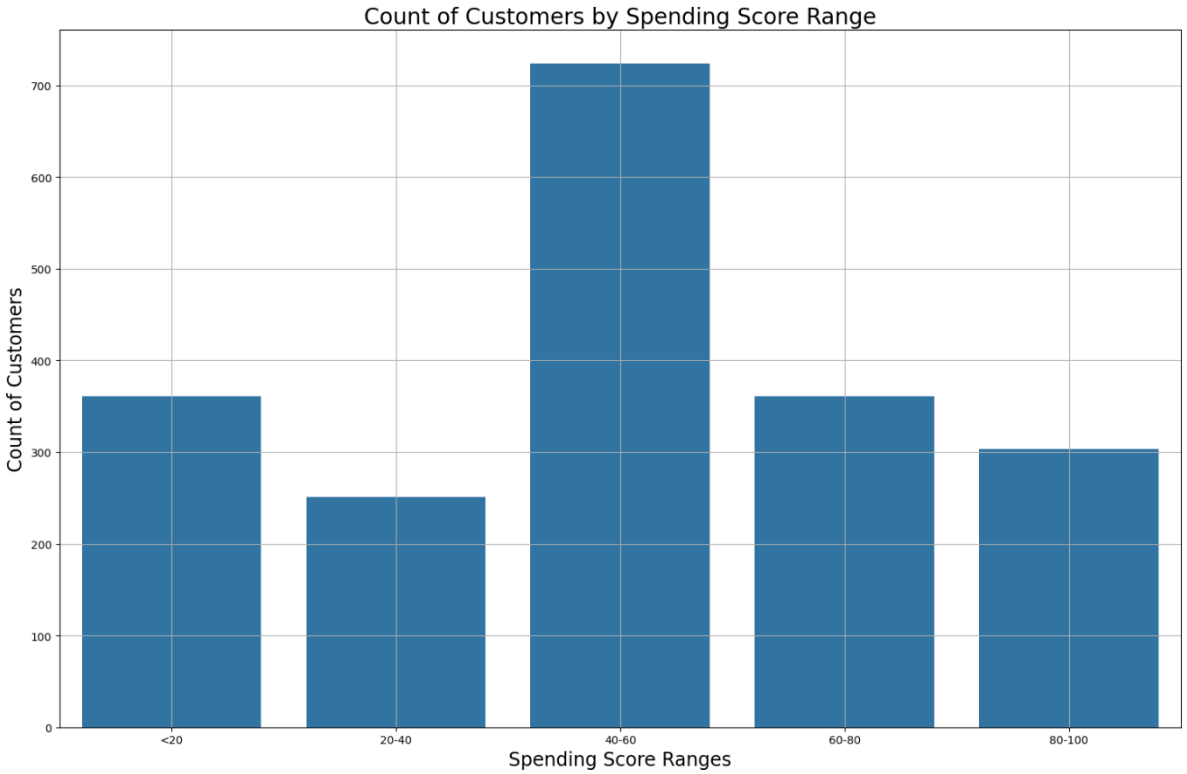



Box Plot of Spending Score
0
20
40
60
80
100
spending_score

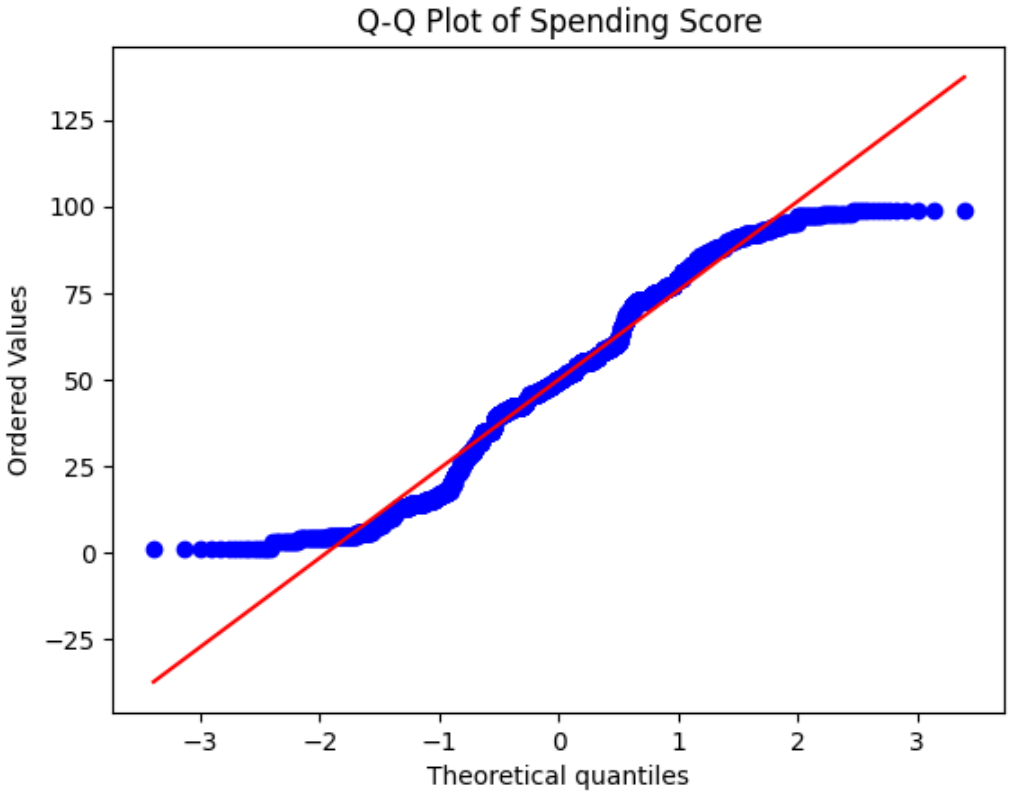
n=2000


Scatter Plot of Spending Score Frequency
Frequency
10
20
30
40
50
60
70
0
20
40
60
80
100
Spending Score

n=2000


Count of Customers by Spending Score Range
Count of Customers
0
100
200
300
400
500
600
700
<20
20-40
40-60
60-80
80-100
Spending Score Ranges

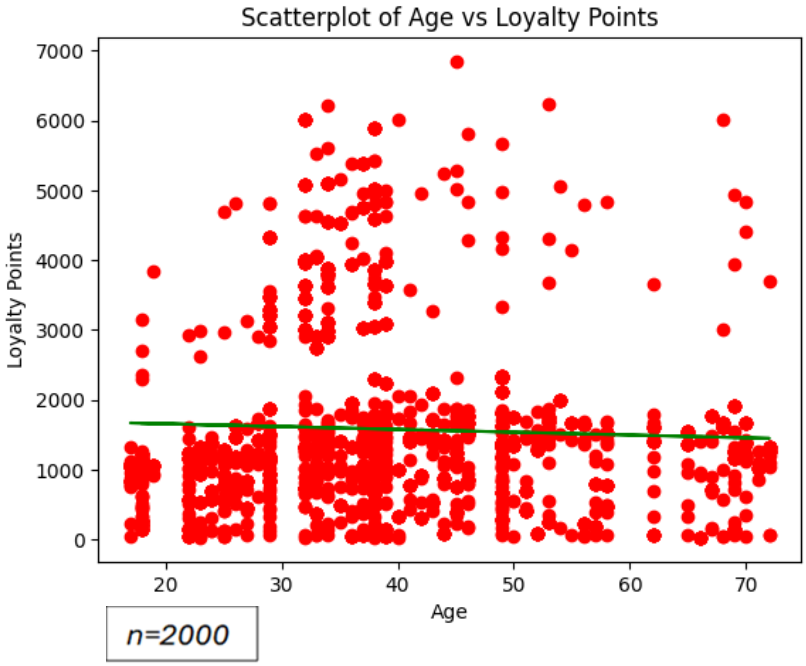
n=2000

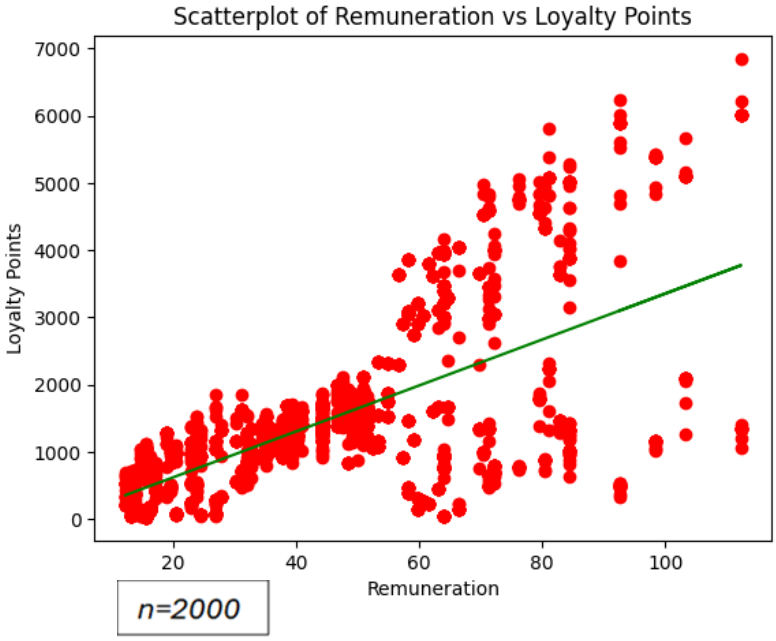

Q-Q Plot of Spending Score
Ordered Values
125
100
75
50
25
0
−25
−3
−2
−1
0
1
2
3
Theoretical quantiles

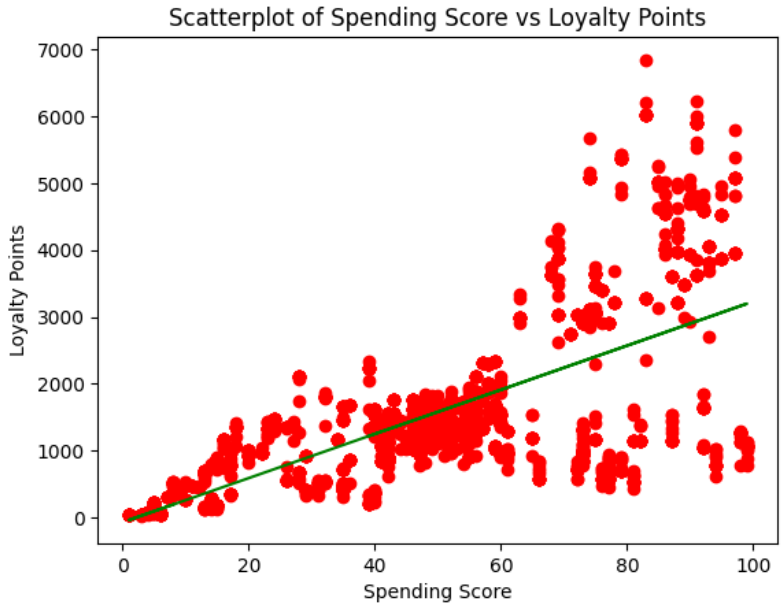
- Q-Q Plot indicates presence of outliers at extremes. Middle range of spending score values however align with normal distribution.

Shapiro-Wilk Test *p-value*=1.453e-20 indicates greater normality of distribution than all other variables, however, an abnormal distribution, nonetheless.

Scatterplots of loyalty points and predictor variables







n=2000

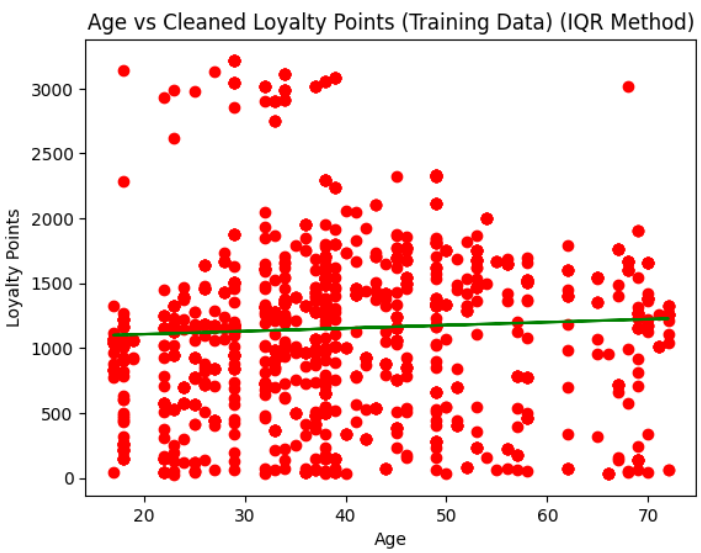
Scatterplot analysis:

Model	ρ
Model 1	-0.42
Model 2	0.616
Model 3	0.616

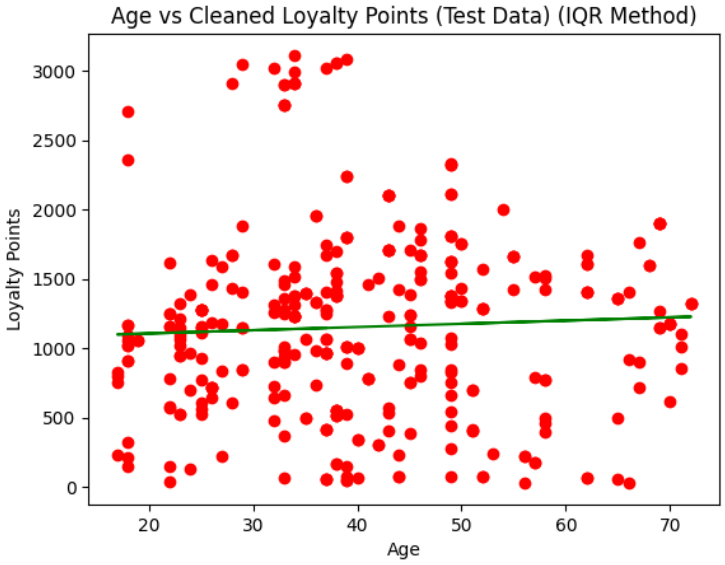
- Model 1 ρ indicates a weak negative linear relationship between age and loyalty points.
- ρ of Model 2 and Model 3 indicate a positive linear relationship. A linear regression model is suitable for capturing the relationships between loyalty points and remuneration, and spending score.

Model 1 (v2)

I recreated my simple regression model in Model 1 but accounted for loyalty point outliers using the IQR method (removing outliers outside the upper and lower bounds). I employed this method due to the highly skewed distribution of loyalty point data.



n=1387



n=345

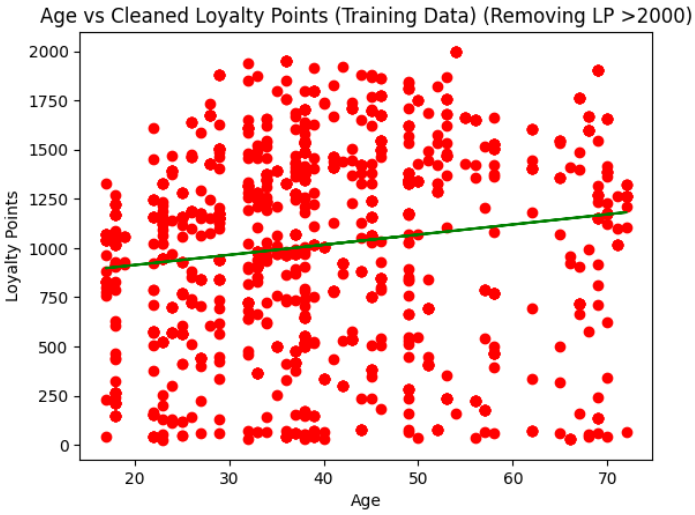
Prediction Metrics (test set):

Model	R-Squared	MAE	MSE	RMSE
Model 1 (v2)	-0.0009	510.12	450405	671

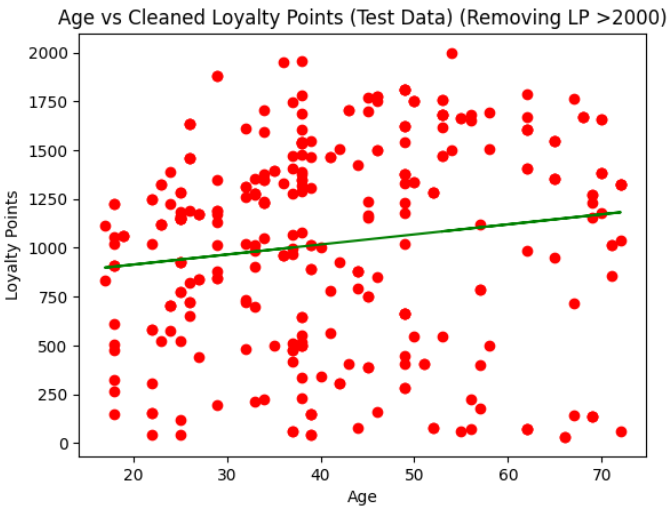
- Model 1(v2) fared with better prediction metrics, indicating a better predictive accuracy of the model, than when raw data was used.
- R-Squared is negative, which suggests that removing such a large contingent of the data removed patterns within said data, which decreased the R-Squared value.
- Overall, Model 1 (v2) lacks any real predictive power, and does not capture any patterns/trends between age and loyalty points.

Model 1 (v3)

Given that I had identified two separate sub-populations within the loyalty point set, I decided to see if there were any patterns in those customers loyalty points below 2000.



n=1387



n=345

Prediction Metrics (test set):

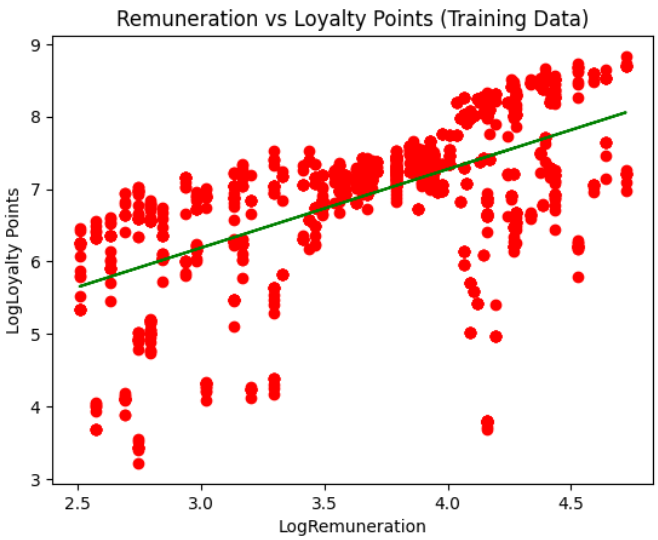
Model	R-Squared	MAE	MSE	RMSE
Model 1 (v3)	0.147	429.657	260847.741	510.733

- R-Squared is now positive, however is still very low. This model only explains 1.47% of variance in loyalty point accumulation.
- Other prediction metrics score better than Model 1 and Model 1 (v2), yet there is still significant prediction error.

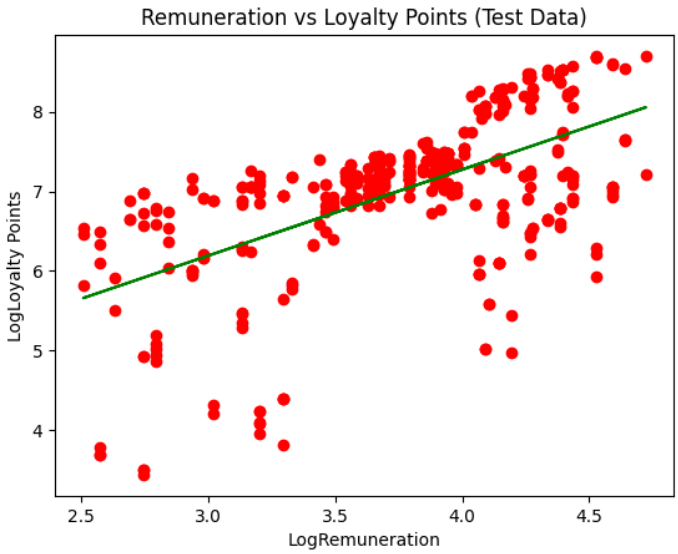
I conclude that a simple linear regression model between age (X), and loyalty points (Y) is unsuitable to capture any relationship, if any, between the two.

Model 2 (v2)

I experimented with using a log transformation on both variables, given the skewed distribution of both. My intention was to see if this would improve the predictive accuracy of my model.



n=1600



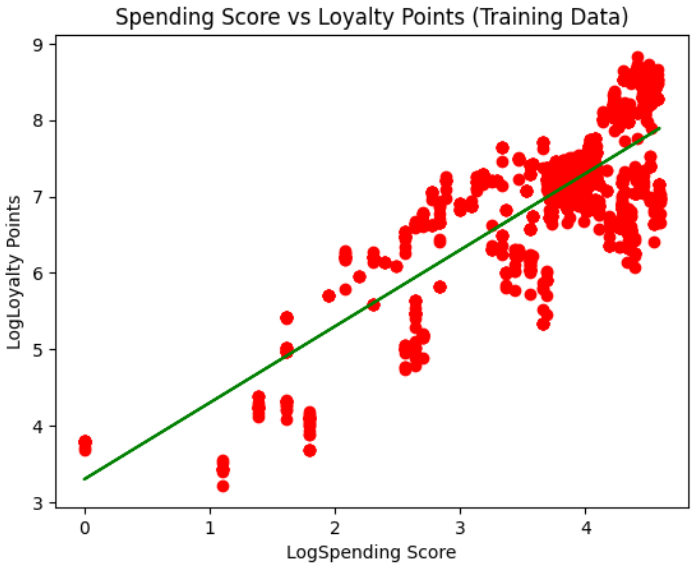
n=400

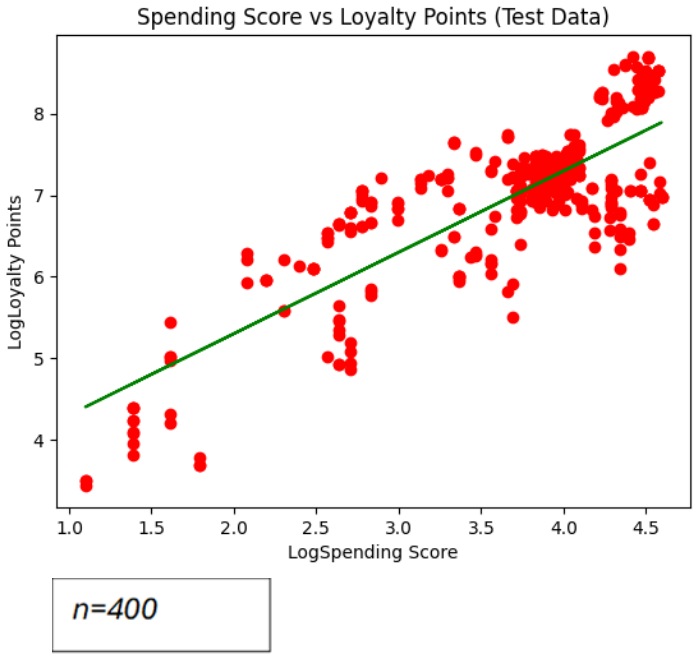
Prediction Metrics (test set):

Model	R-Squared	MAE	MSE	RMSE
Model 2 (v2)	0.34	768.202	1165917.033	1051.7

- Model 2 (v3) had a marginally higher R-Squared, indicating the model captured slightly more variance in loyalty point scores.
- However, the other prediction metrics were higher, indicating less model accuracy.
- I theorise that these results are due to the wide range of loyalty point values. Upon transforming the log-transformed data back to original scale, the model is worse in its predictive abilities due to this wide range.

Model 3 (v2)





Prediction Metrics (test set):

Model	R-Squared	MAE	MSE	RMSE
Model 3 (v2)	0.664	652.954	943522.424	971.351

- A higher R-Squared than when using raw data indicates a stronger fit on the test data.
- However, we are presented with a similar case as in Model 2 (v2), where we encounter greater prediction error.

Summary

- A simple linear regression model between age (X) and loyalty points (Y) is unsuitable. There is no real relationship between the two.
- A simple linear regression model to capture the relation between remuneration (X) and loyalty points (Y) is suitable. We see however, that the abnormal distribution of loyalty point values reduces the predictive power of a model. The same findings can be said when X is spending score.
- Spending score was most positively correlated with loyalty point accumulation, closely followed by remuneration.
- Customers can be divided into two sub-populations based on their loyalty points (>2000 and <2000 points).

Q2. How well can decision trees be used to understand loyalty point accumulation?

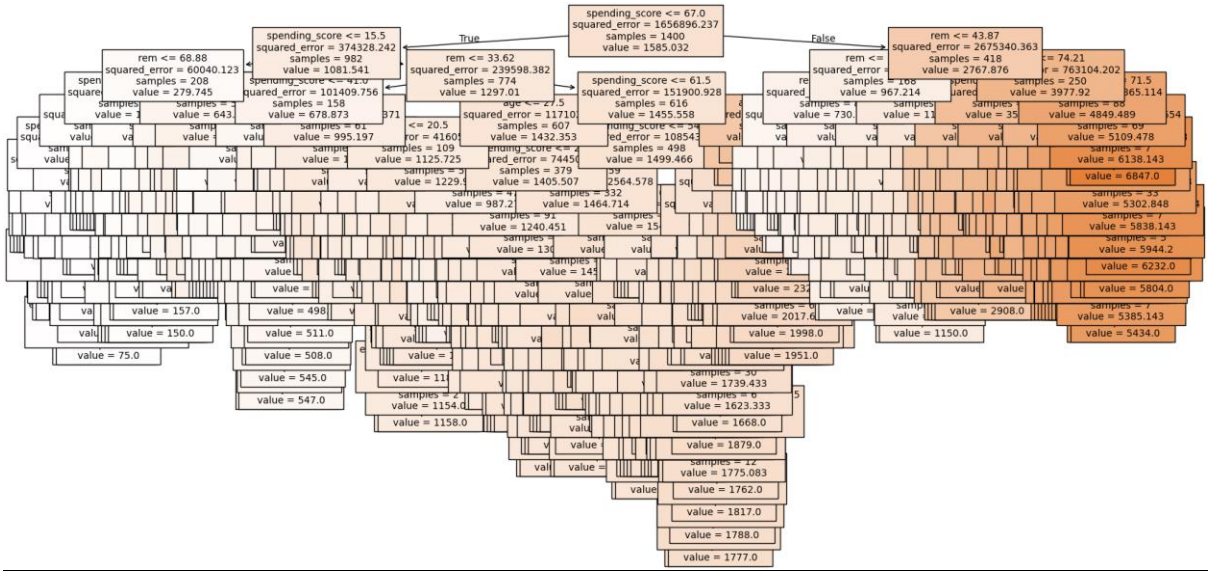
Building a decision tree to explore loyalty point accumulation was an exercise in balancing overfitting the model (reducing interpretability) and creating an interpretable output (increasing prediction error).

I focused on using a decision tree regressor model whereby:

$$y = f(x1, x2, x3, ..., xn)$$

$$loyaltypoints = f(age, remuneration, spendingscore, gender, education)$$

Initial model creation



Model	MAE	MSE	RMSE
Tree 1	31.95	7599.02	87.17

- The model is overly complex and impractical.
- Prediction metrics indicate a moderate degree of accuracy.

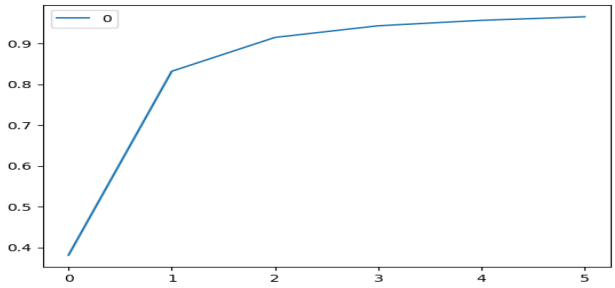
Pre-pruning

I employed pre-pruning to find a prime balance between interpretability and accuracy.

```
max_depth_range = list(range(1, 7))

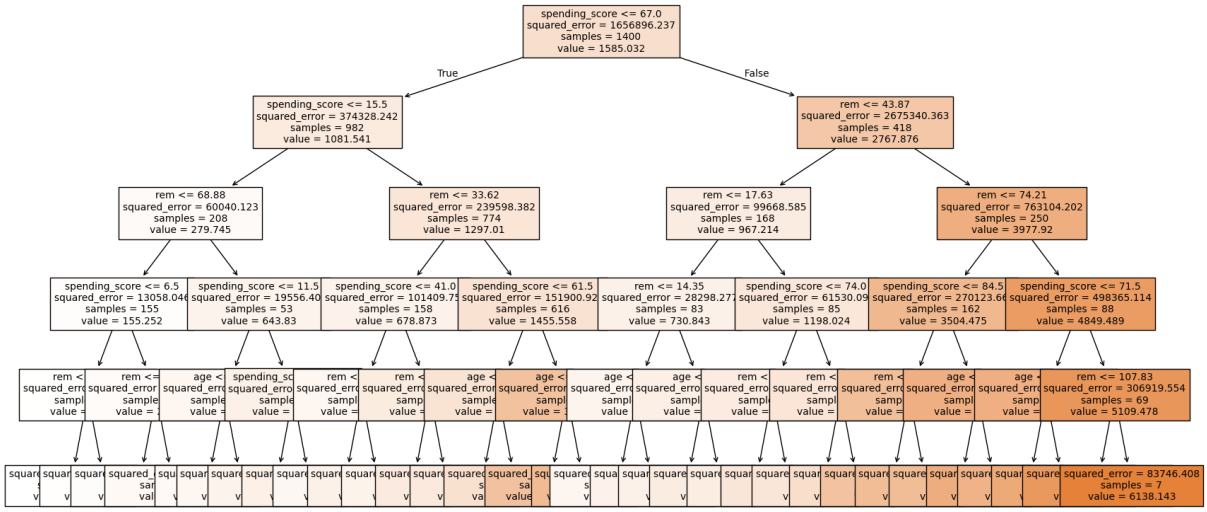
accuracy = []
for depth in max_depth_range:
    dtr = DecisionTreeRegressor(max_depth = depth, random_state = 10)
    dtr.fit(X_train, y_train)
    score = dtr.score(X_test, y_test)
    accuracy.append(score)

accuracy = pd.DataFrame(accuracy)
accuracy.plot()
```



- Based on this code, the most accurate maximum depth should be 5.

I remade my tree specifying this parameter.



Model	MAE	MSE	RMSE
Tree 1 (v2)	176.81	70121.16	264.8

- While the tree is now much more readable, significant prediction error persists.

In trying to find the best parameters, I utilised GridSearchCV, to find the optimum parameters.

```python
regressor = DecisionTreeRegressor(random_state=42)
```

```python
grid_search = GridSearchCV(estimator=regressor,
                           param_grid=param_grid,
                           cv=5, n_jobs=-1, verbose=2,
                           scoring='neg_mean_squared_error')
```

```python
grid_search.fit(X_train, y_train)

best_regressor = grid_search.best_estimator_

y_pred = best_regressor.predict(X_test)

mse = mean_squared_error(y_test, y_pred)

rmse = mse ** 0.5

best_params = grid_search.best_params_

print(f"Best parameters: {best_params}")
print(f"Test RMSE: {rmse}")
```

```
Fitting 5 folds for each of 48600 candidates, totalling 243000 fits
Best parameters: {'max_depth': 6, 'min_samples_leaf': 10, 'min_samples_split': 26}
Test RMSE: 203.8609359164831
```

Tree with optimum parameters

Note: Due to the size of the output, view the tree here:

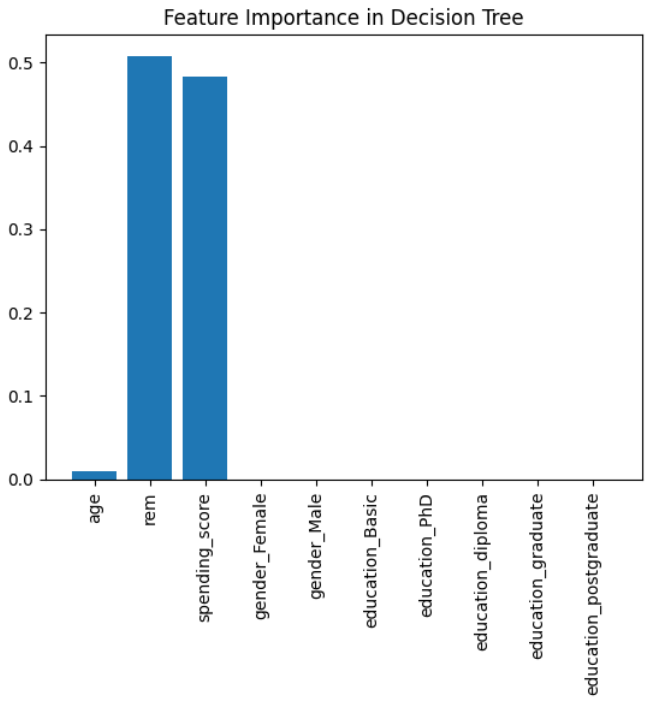
https://github.com/amanlang88888/amanlangprojectarchive/blob/main/decisiontreeregressoroptimal.png

Model	MAE	MSE	RMSE
Tree 1(v3)	139.69	41559.28	203.87

- The tree is somewhat readable and gives us somewhat better prediction accuracy as opposed to Tree 1(v2).
- However, while the model is no longer heavily overfitted, it is difficult and cumbersome to interpret the model and visually intake any patterns/trends.

Feature importance

Another approach I implemented was to only use statistically significant predictor variables. I did this to reduce model complexity.



- It is evident that remuneration, spending score, and, to a much lesser extent, age are significant variables.

Combining pruning techniques

I remade my model focusing only on customers earning <2000 loyalty points and using only remuneration and spending score. I also employed GridSearchCV to find optimal parameters.

https://github.com/amanlang88888/amanlangprojectarchive/blob/main/treegoodunder2000.png

Model	MAE	MSE	RMSE
Tree 1(v4)	119.72	26800.53	163.71

- Prediction metrics indicate significant prediction error.
- However, the model is still interpretable.
- While the prediction metrics do not indicate as good a fit as when all features were used, and no parameters were specified, the model is more readable.
- The trend shown is that generally, positive increase in both remuneration and spending score is associated with greater loyalty point accumulation.

Summary

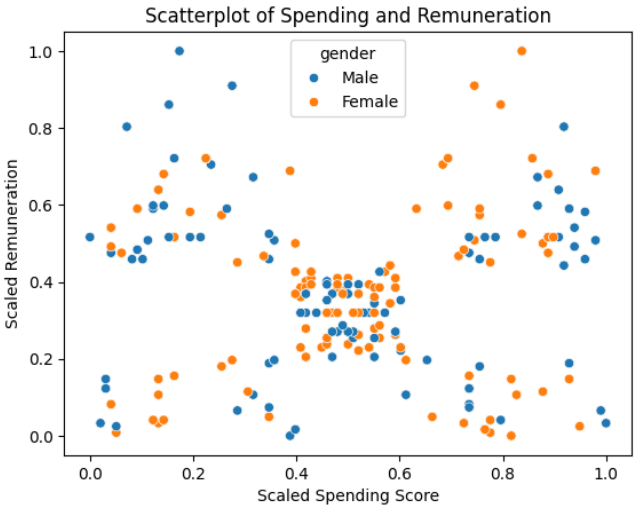
- Remuneration and spending score are key determinants in predicting loyalty points, presenting positive correlation in both variables with respect to loyalty points.
- A decision tree regressor model can be used in the context of predicting loyalty point accumulation, however it presents drawbacks:
    - Pre-pruning reduces overfitting, while reducing accuracy.
    - The continuous nature of the variables means that insights are difficult to present.

Q3. Can k-means clustering aid in understanding trends within remuneration and spending score?

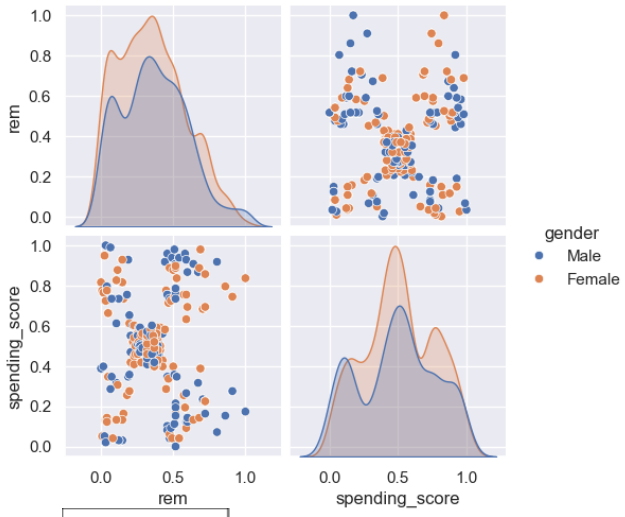
I wanted to explore if there are any correlations or groups when looking at remuneration and spending score.

Initial exploration

I wanted to see if there were any patterns with remuneration and spending score in interactions with gender. I scaled my data using Min-Max method, due to the lack of extreme outliers, and abnormal distribution.



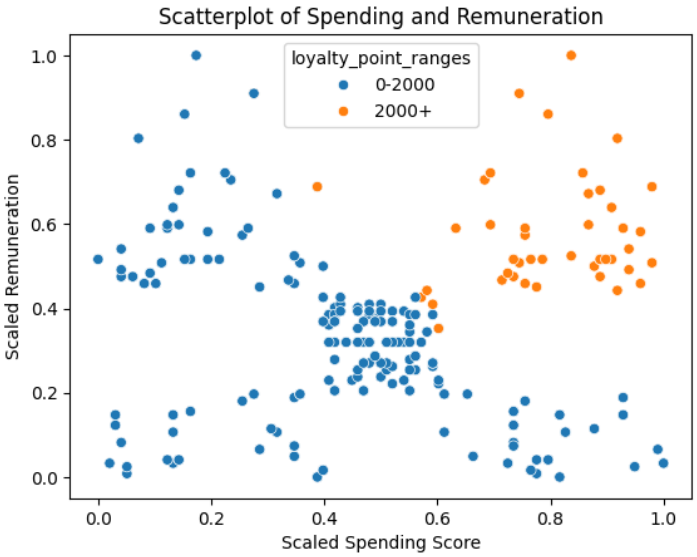
n=2000



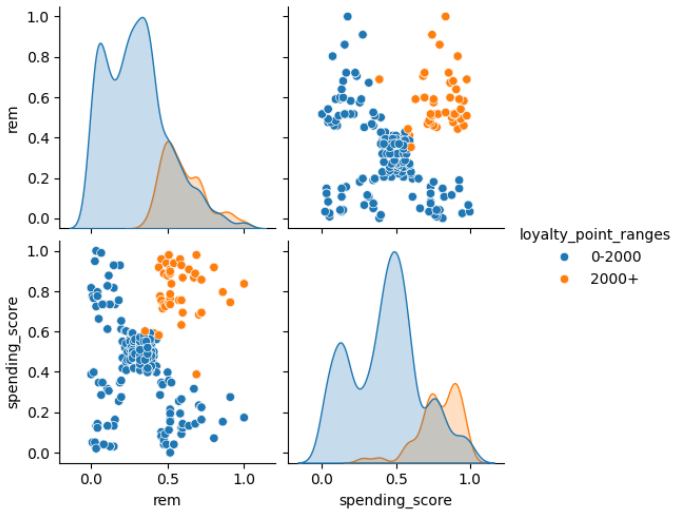
n=2000

- The scatterplot does not show us any discerning trends. The distribution of gender between remuneration and spending score appears random.
- The pairgrid however shows us that males will mirror females in remuneration and spending score values, however, their scores are significantly lower.

I was curious to see interactions with loyalty points.



n=2000



n=2000

- There is a more evident trend here. High loyalty point earners (>2000 points) are associated with higher remuneration and spending scores.
- Those in the >2000 range are a distinct population from those in the 0-2000 range.

Finding the optimal number of clusters

I find the optimal number of clusters, *k*, to ensure meaning grouping between remuneration and spending score, whereby:

$$k = f(x1, x2)$$
$$k = f(remuneration, spendingscore)$$

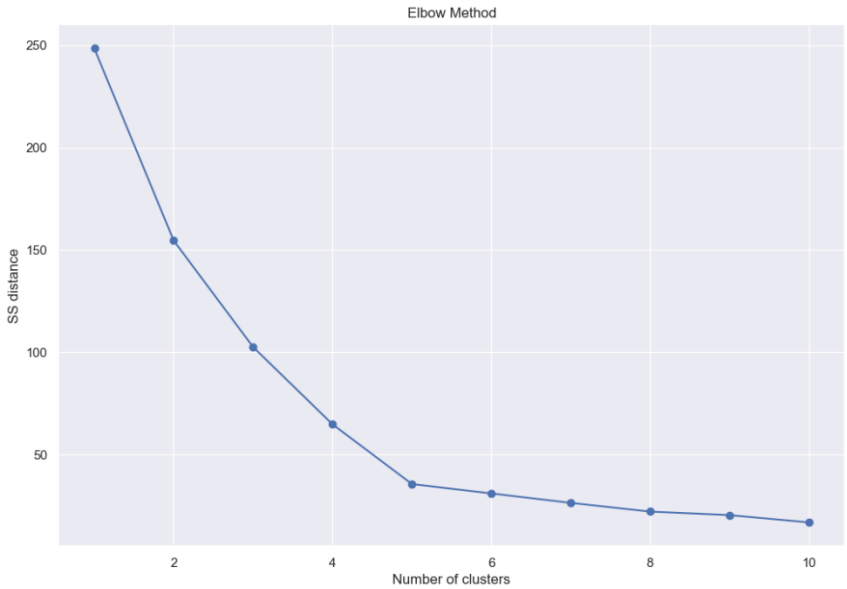
I used the Elbow Method to determine *k*.

```
x = dfgender[['rem', 'spending_score']]
ss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i,
                    init='k-means++',
                    max_iter=300,
                    n_init=10,
                    random_state=0)
    kmeans.fit(x)
    ss.append(kmeans.inertia_)

plt.plot(range(1, 11),
         ss,
         marker='o')

plt.title("Elbow Method")
plt.xlabel("Number of clusters")
plt.ylabel("SS distance")
plt.show()
```



- *k=5* according to this method.

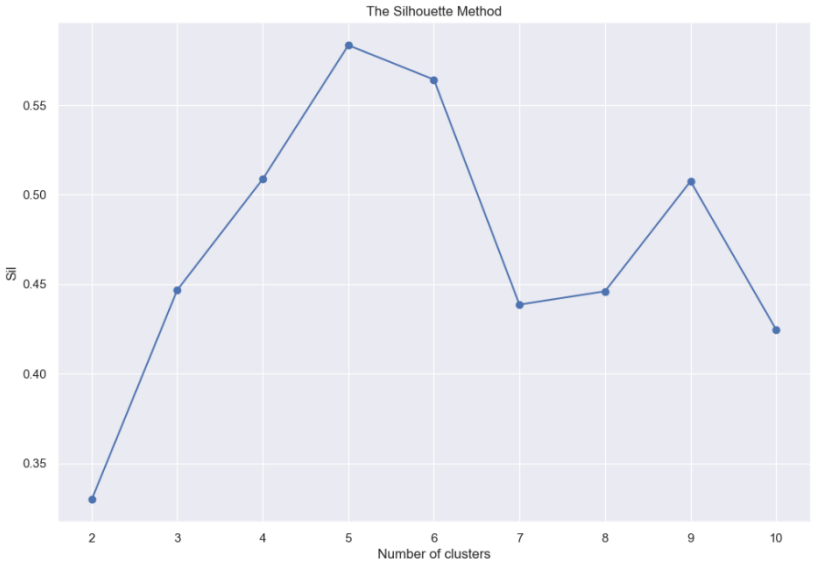
I used the Silhouette Method to find silhouette score for different *k* numbers.

```
x = dfgender[['rem', 'spending_score']]

sil = []
kmax = 10

for k in range(2, kmax+1):
    kmeans_s = KMeans(n_clusters=k).fit(x)
    labels = kmeans_s.labels_
    sil.append(silhouette_score(x,
                                labels,
                                metric='euclidean'))

plt.plot(range(2, kmax+1),
         sil,
         marker='o')
plt.title("The Silhouette Method")
plt.xlabel("Number of clusters")
plt.ylabel("Sil")
plt.show()
```



- *k*=5 would garner the highest silhouette score.

I also implemented another method, which also calculates silhouette score and coefficient, yet provides cluster visualisation for each *k* number.

```
x = dfgender[['rem', 'spending_score']].to_numpy()
range_n_clusters = range(2, 11)

for n_clusters in range_n_clusters:
    fig, (ax1, ax2) = plt.subplots(1, 2)
    fig.set_size_inches(18, 7)
    ax1.set_xlim([-0.1, 1])
    ax1.set_ylim([0, len(x) + (n_clusters + 1) * 10])
    clusterer = KMeans(n_clusters=n_clusters, random_state=10)
    cluster_labels = clusterer.fit_predict(x)

    silhouette_avg = silhouette_score(x, cluster_labels)
    print(
        "For n_clusters =",
        n_clusters,
        "The average silhouette_score is :",
        silhouette_avg,
    )

    sample_silhouette_values = silhouette_samples(x, cluster_labels)

    y_lower = 10
    for i in range(n_clusters):
        ith_cluster_silhouette_values = sample_silhouette_values[cluster_labels == i]

        ith_cluster_silhouette_values.sort()

        size_cluster_i = ith_cluster_silhouette_values.shape[0]
        y_upper = y_lower + size_cluster_i

        color = cm.nipy_spectral(float(i) / n_clusters)
        ax1.fill_betweenx(
            np.arange(y_lower, y_upper),
            0,
            ith_cluster_silhouette_values,
            facecolor=color,
            edgecolor=color,
            alpha=0.7,
        )

        ax1.text(-0.05, y_lower + 0.5 * size_cluster_i, str(i))

        y_lower = y_upper + 10

    ax1.set_title("The silhouette plot for the various clusters.")
    ax1.set_xlabel("The silhouette coefficient values")
    ax1.set_ylabel("Cluster label")
    ax1.axvline(x=silhouette_avg, color="red", linestyle="--")
    ax1.set_yticks([])
    ax1.set_xticks([-0.1, 0, 0.2, 0.4, 0.6, 0.8, 1])

    colors = cm.nipy_spectral(cluster_labels.astype(float) / n_clusters)
    ax2.scatter(
        x[:, 0], x[:, 1], marker=".", s=30, lw=0, alpha=0.7, c=colors, edgecolor="k"
    )

    centers = clusterer.cluster_centers_
    ax2.scatter(
        centers[:, 0],
        centers[:, 1],
        marker="o",
        c="white",
        alpha=1,
        s=200,
        edgecolor="k",
    )

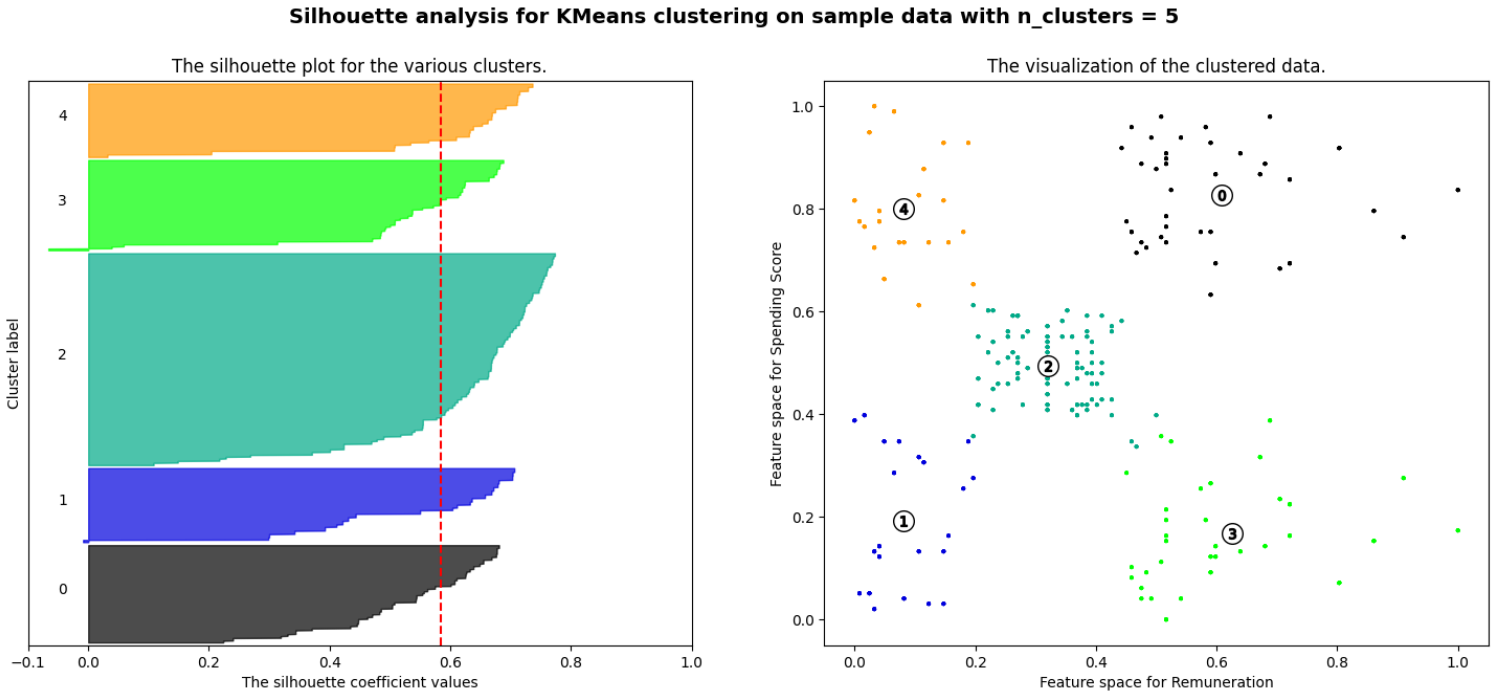
    for i, c in enumerate(centers):
        ax2.scatter(c[0], c[1], marker="$%d$" % i, alpha=1, s=50, edgecolor="k")

    ax2.set_title("The visualization of the clustered data.")
    ax2.set_xlabel("Feature space for the 1st feature")
    ax2.set_ylabel("Feature space for the 2nd feature")

    plt.suptitle(
        "Silhouette analysis for KMeans clustering on sample data with n_clusters = %d"
        % n_clusters,
        fontsize=14,
        fontweight="bold",
    )

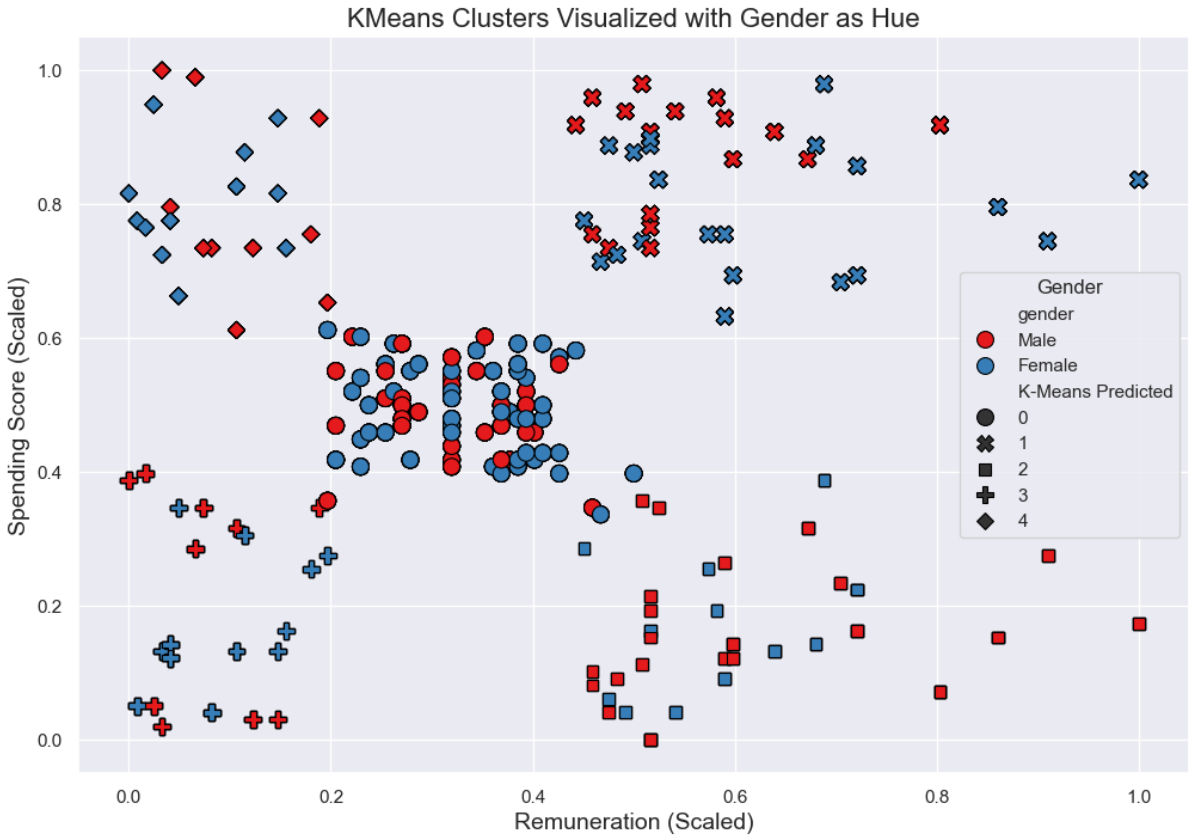
plt.show()
```

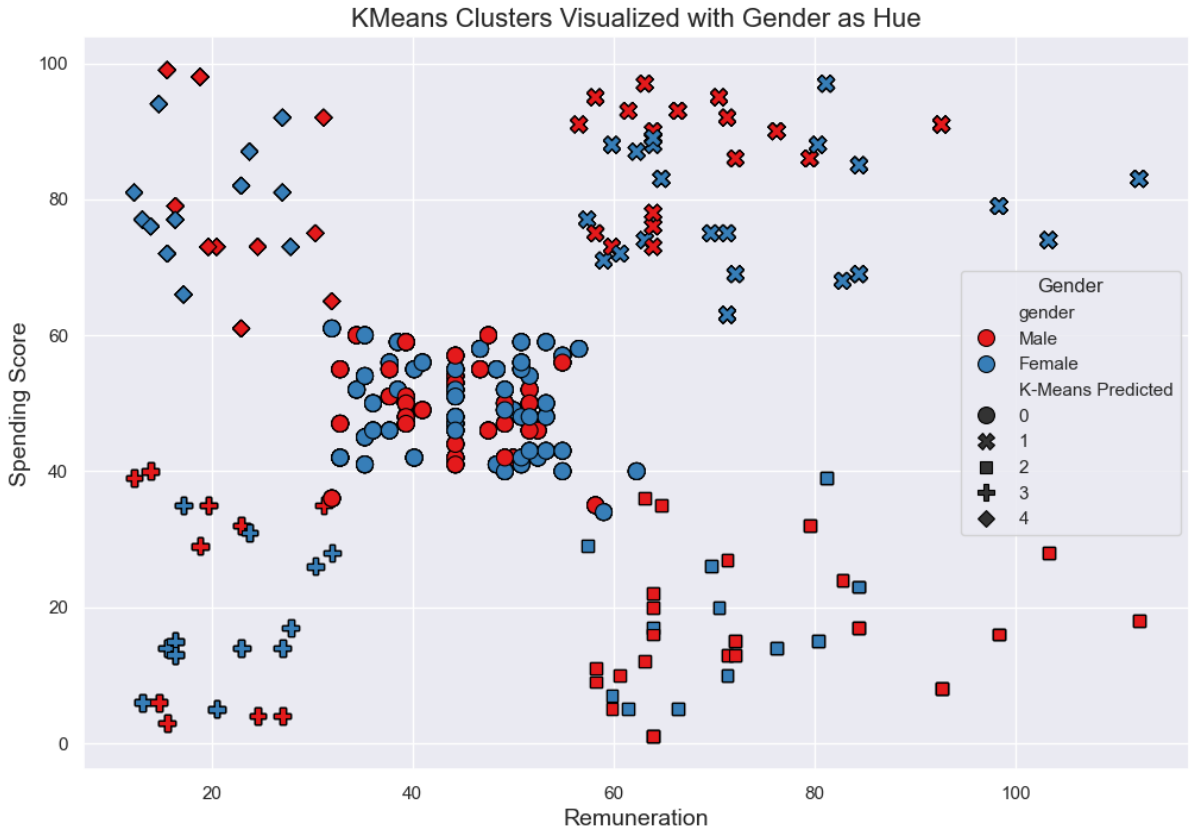
```
For n_clusters = 2 The average silhouette_score is : 0.3398674428838815
For n_clusters = 3 The average silhouette_score is : 0.4467670819091451
For n_clusters = 4 The average silhouette_score is : 0.511792405732587
For n_clusters = 5 The average silhouette_score is : 0.5834518038964438
For n_clusters = 6 The average silhouette_score is : 0.5372779302420929
For n_clusters = 7 The average silhouette_score is : 0.5165407672155633
For n_clusters = 8 The average silhouette_score is : 0.4705423336564262
For n_clusters = 9 The average silhouette_score is : 0.4760223535494646
For n_clusters = 10 The average silhouette_score is : 0.46942231482532837
```



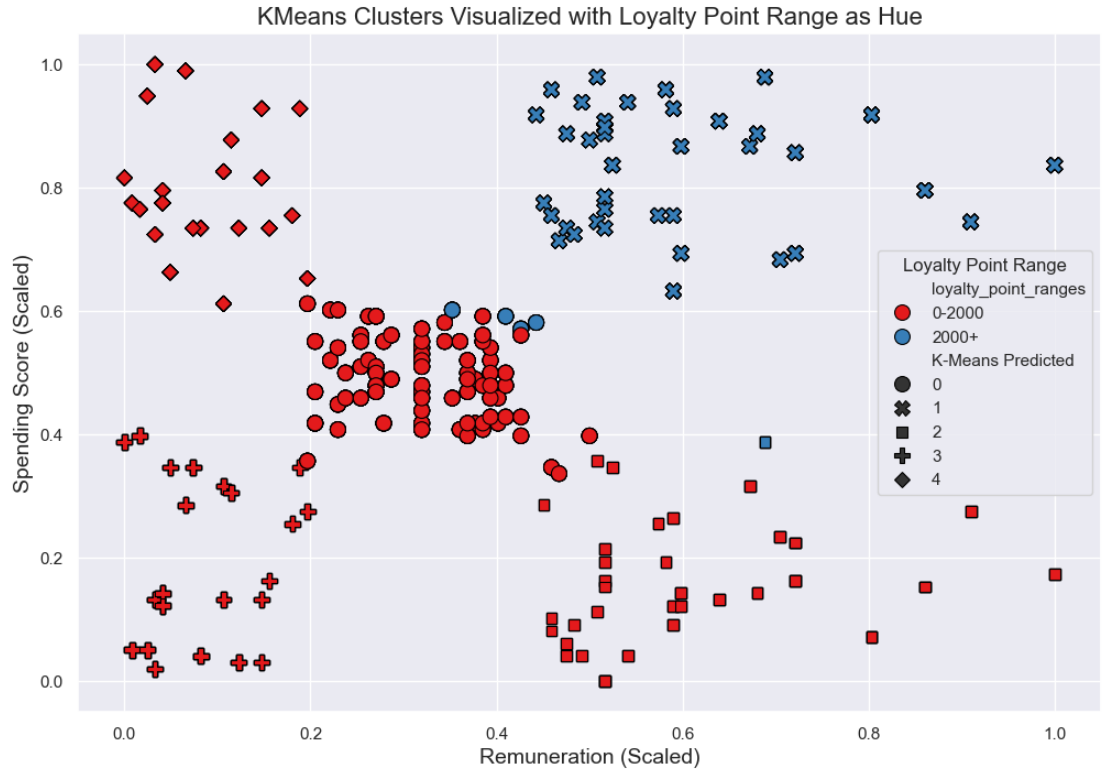
- We can confirm *k*=5 allows for meaningful and distinct clusters.

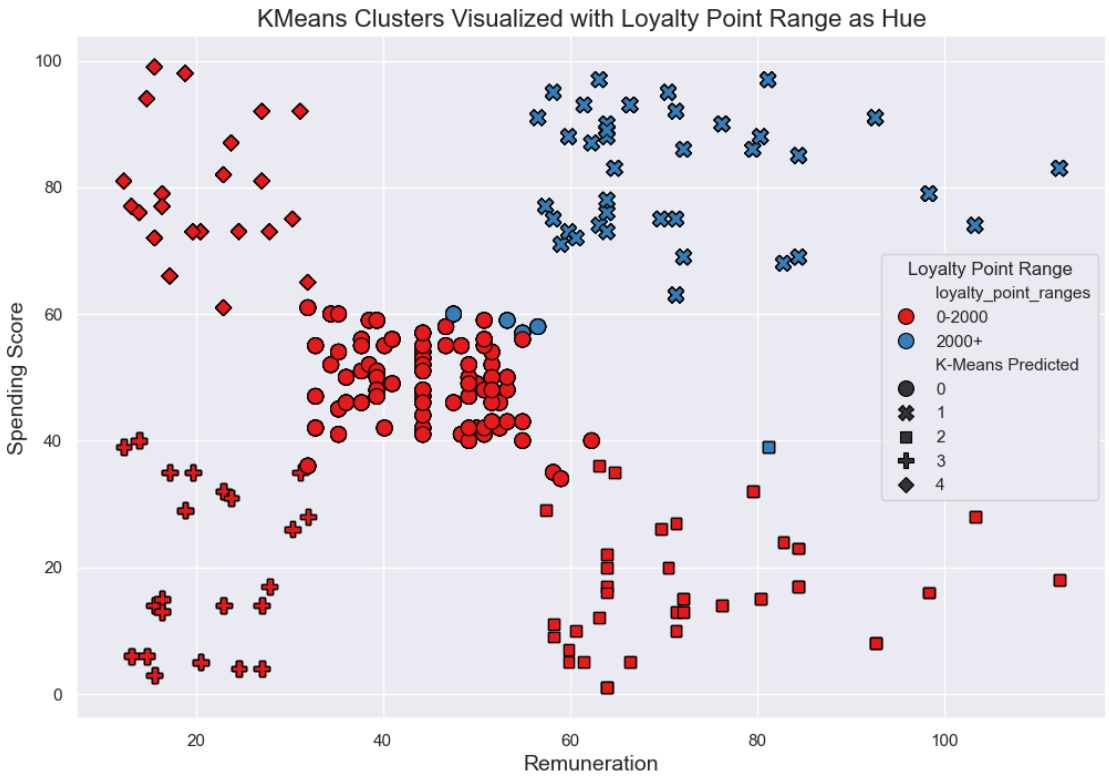
Visualizations





- We see that there is not much of a pattern when looking at how gender interacts with *k*-means predicted clusters.





- We see one cluster is separate, and this largely is the >2000 points population, which is distinct from all other clusters. Customers in this cluster have high remuneration and spending score.

Summary

- There are five distinct groups of customers with similar remuneration and spending scores.
- There is not much of a distinct pattern of gender within these groups.
- High loyalty point earners are largely a distinct group from the rest of the population.

Q4

I decided to focus on only using remuneration and spending score as my predictor variables, being the most statistically significant in relation to loyalty points.

I created a multiple linear regression model whereby:

$$y = \beta 0 + \beta 1 x1 + \beta 2 x2 + \epsilon$$

$$loyaltypoints = \beta 0 + \beta 1 \cdot remuneration + \beta 2 \cdot spendingscore + \epsilon$$

Initial model creation

I created an initial model using raw data.

```
> model <- lm(loyalty_points ~ rem + spending_score,
+             data = Subset)
> summary(model)

Call:
lm(formula = loyalty_points ~ rem + spending_score, data = Subset)

Residuals:
     Min       1Q   Median       3Q      Max
-1646.02  -363.66    40.34   280.59  1999.95

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1700.3051    35.7396  -47.58   <2e-16 ***
rem               33.9795     0.5166   65.77   <2e-16 ***
spending_score    32.8927     0.4578   71.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 534.1 on 1997 degrees of freedom
Multiple R-squared:  0.8269,    Adjusted R-squared:  0.8267
F-statistic:  4770 on 2 and 1997 DF,  p-value: < 2.2e-16

>
> model_summ <- summary(model)
> mean(model_summ$residuals^2)
[1] 284879.8
```

- R-Squared, F-Statistic indicates model is a strong fit on the data.
- There is a strong relationship between the outcome and predictor variables.
- However, the model is not very accurate, due to a high MSE.

Assumptions for MLR

I conducted exploratory analytics.

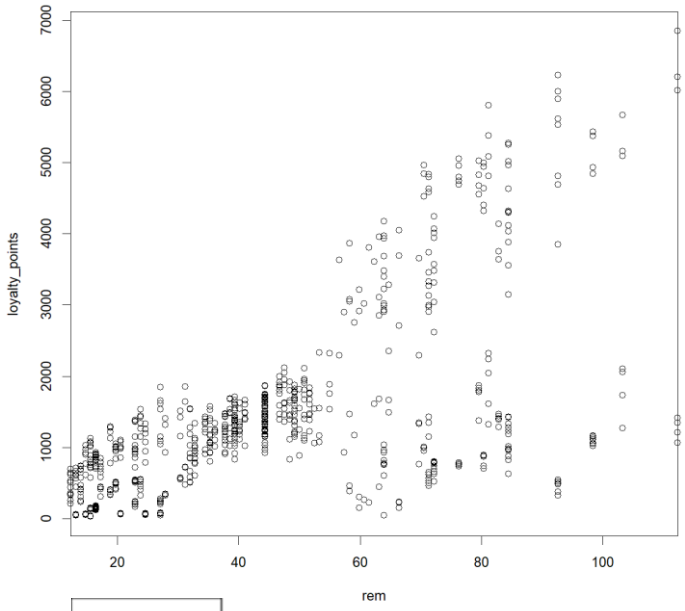
1. **Multicollinearity:**

```
> cor(Subset$rem, Subset$spending_score)
[1] 0.005612492

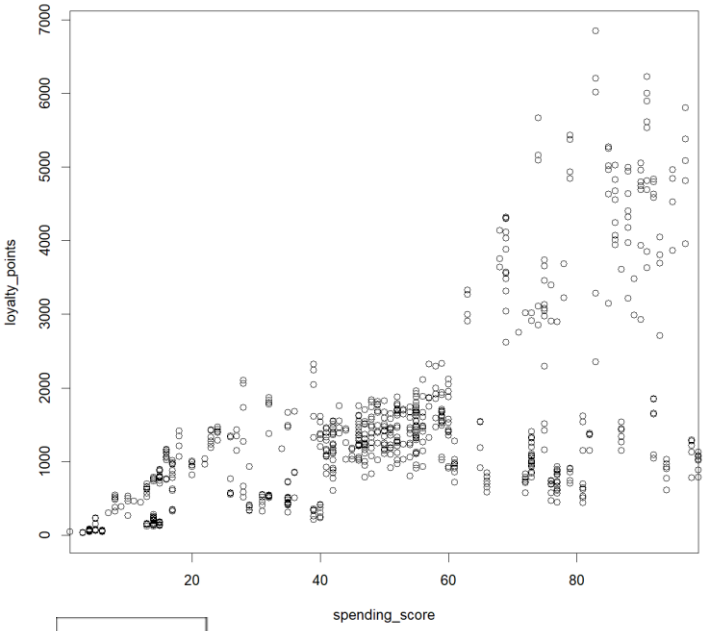
> vif(model)
            rem spending_score
       1.000032       1.000032
```

- There is low correlation between remuneration and spending score.
- Low VIF values indicate little multicollinearity.

2. **Linearity:**



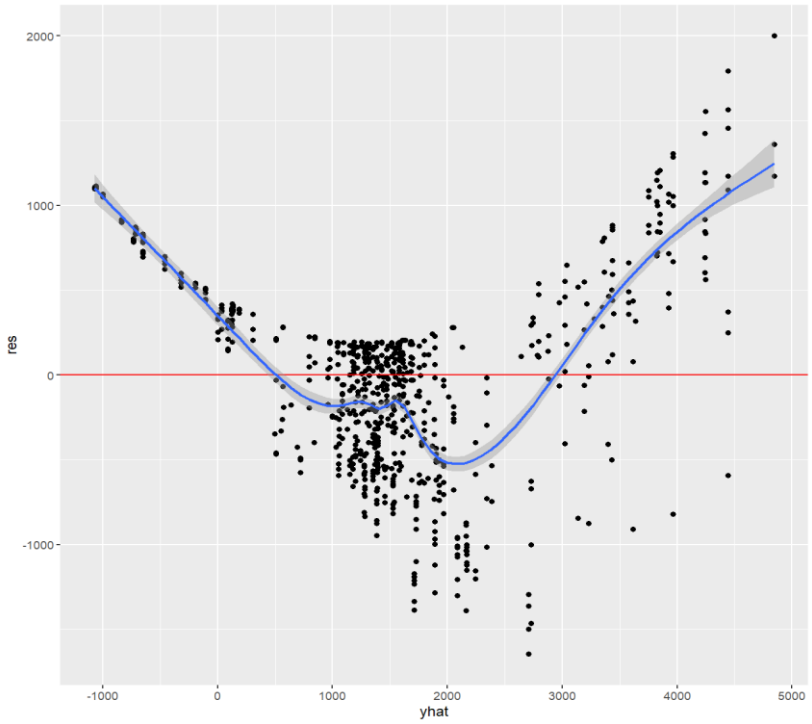
n=2000



n=2000

- Visually, Y~X1, Y~X2 is linear.

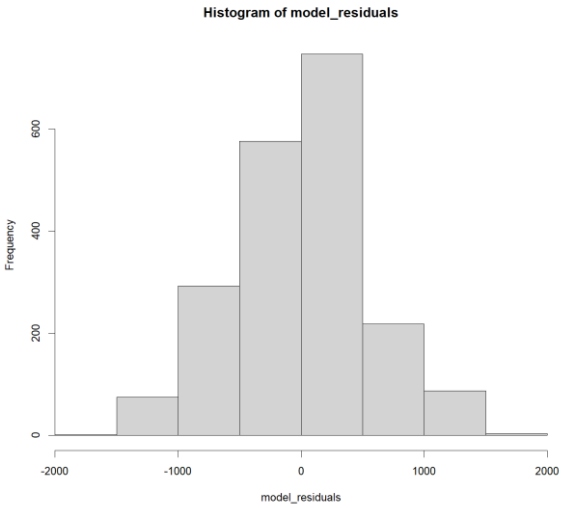
3. Independence of Residuals

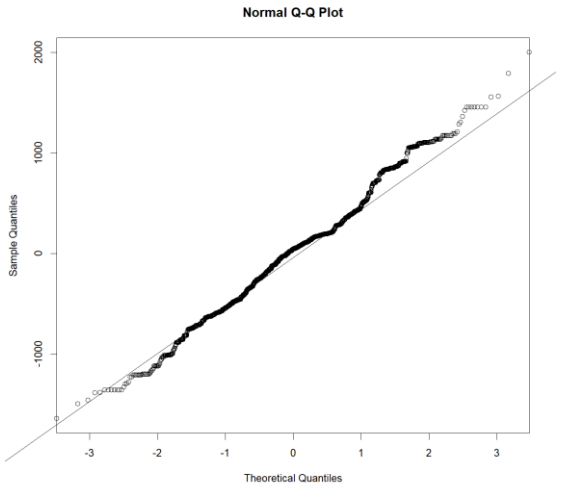


- From here, we see that our residuals do not align to predicted values.
- The curve in the plotted residuals suggest linearity is violated. This may be due to non-linear patterns in residuals.

4. Normality

As previously established, loyalty point values are abnormally distributed. I check for normality of residual distribution in Y.





- The histogram would suggest, visually a more normal distribution of residuals.
- Q-Q Plot suggests that there are heavy outliers at the right tail.

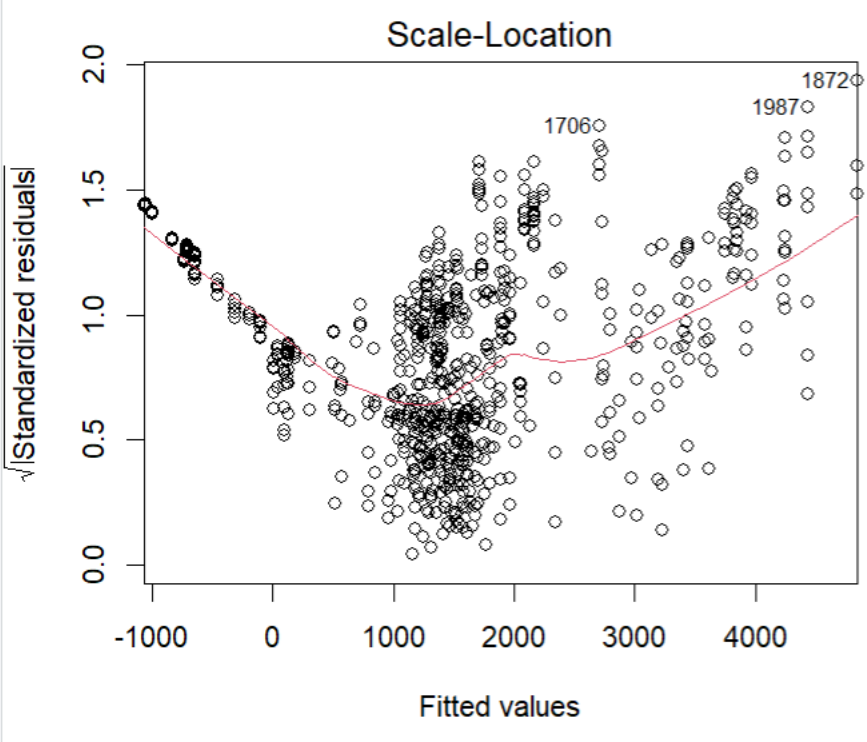
```
> kurtosis(model_residuals)
[1] 3.115447
```

```
        Shapiro-Wilk normality test

data:  model_residuals
W = 0.99158, p-value = 2.391e-09
```

- The residuals are abnormally distributed.
- There are heavy right tails.
- The residuals are asymmetrically distributed.

5. **<u>Homoscedasticity</u>**



- Variance of standardised residuals is not consistent across the range of fitted values. This indicates heteroscedasticity.

Polynomial regression

Due to issues with abnormal residual distribution, heteroscedasticity, and non-linearity of residuals, I implement polynomial regression with the intention of capturing possible non-linear patterns, whereby:

$$y = \beta 0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \ldots + \beta_h x_1^h + \beta_{h+1} x_2^h + \epsilon$$

$$loyaltypoints = \beta 0 + \beta_1 \cdot remuneration + \beta_2 \cdot spendingscore + \beta_3 \cdot remuneration^2 + \beta_4 \cdot spendingscore^2 + \beta_5 \cdot remuneration \cdot spendingscore \ldots + \beta_h \cdot remuneration^h + \beta_{h+1} \cdot spendingscore^h + \epsilon$$

I want to find the optimal degree, *h*, to strike a balance between model flexibility in capturing complex patterns, whilst preventing overfitting. To do this, I performed k-fold cross validation, and calculated MSE for each *h* value. I set *k*=10, in consideration of computational efficiency, and as a bias-variance balance.

```
# Randomly shuffle data:
Subset.shuffled <- Subset[sample(nrow(Subset)),]

# Define no. of folds to use for k-fold cross-validation:
K <- 10

# Define degree of polynomials to fit.
degree <- 5

# Create k equal-sized folds:
folds <- cut(seq(1,nrow(Subset.shuffled)),breaks=K,labels=FALSE)

# Create object to hold model MSE's:
mse = matrix(data=NA,nrow=K,ncol=degree)

# Perform K-fold cross validation:
for(i in 1:K){

  # Define training and testing data:
  testIndexes <- which(folds==i,arr.ind=TRUE)
  testData <- Subset.shuffled[testIndexes, ]
  trainData <- Subset.shuffled[-testIndexes, ]

  # Use k-fold cv to evaluate models:
  for (j in 1:degree){
    fit.train = lm(loyalty_points ~ poly(rem,j) +
                     poly(spending_score,j), data=trainData)
    fit.test = predict(fit.train, newdata=testData)
    mse[i,j] = mean((fit.test-testData$loyalty_points)^2)
  }
}

# Find MSE for each polynomial degree:
colMeans(mse)
print(mse)
```

```
> colMeans(mse)
[1] 286140.5 285827.4 285253.9 286657.9 272476.3
> print(mse)
          [,1]     [,2]     [,3]     [,4]     [,5]
 [1,] 277160.2 275963.0 287156.2 291461.6 292183.4
 [2,] 298104.2 302739.0 301719.2 309367.3 307547.7
 [3,] 270277.8 266815.6 265604.1 261747.0 239738.2
 [4,] 282987.1 285657.6 281119.2 283108.0 254913.0
 [5,] 276695.0 276600.2 272474.1 275332.4 265483.0
 [6,] 291371.2 291459.3 291129.0 290361.7 268871.9
 [7,] 304087.5 299797.7 304226.0 304627.1 299537.7
 [8,] 286792.2 288983.8 284505.2 283439.3 271270.7
 [9,] 285409.1 284025.8 282220.5 286552.8 264993.0
[10,] 288520.5 286231.5 282385.7 280582.3 260224.1
```

- *h*=5 produced the lowest MSE, indicating the optimal *h* value.

```
Call:
lm(formula = loyalty_points ~ poly(rem, 5) + poly(spending_score,
    5), data = Subset)

Residuals:
     Min       1Q   Median       3Q      Max
-1438.76  -331.30    31.29   270.97  1886.74

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                  1578.03      11.57 136.422  < 2e-16 ***
poly(rem, 5)1               35381.10     555.86  63.652  < 2e-16 ***
poly(rem, 5)2                 455.90     638.03   0.715 0.474975
poly(rem, 5)3                -549.13     631.13  -0.870 0.384366
poly(rem, 5)4                -955.17     519.87  -1.837 0.066310 .
poly(rem, 5)5                1002.60     598.94   1.674 0.094293 .
poly(spending_score, 5)1    38385.33     517.42  74.185  < 2e-16 ***
poly(spending_score, 5)2      844.67     720.50   1.172 0.241199
poly(spending_score, 5)3    -1921.80     520.46  -3.693 0.000228 ***
poly(spending_score, 5)4     1699.08     645.33   2.633 0.008532 **
poly(spending_score, 5)5    -5405.83     518.85 -10.419  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 517.3 on 1989 degrees of freedom
Multiple R-squared:  0.8383,    Adjusted R-squared:  0.8375
F-statistic:  1031 on 10 and 1989 DF,  p-value: < 2.2e-16

>
> poly_model_summ <- summary(poly_model)
> mean(poly_model_summ$residuals^2)
[1] 266132.7
```

- R-Squared value indicates that the polynomial regression model explains a large portion of variance in loyalty points.
- MSE is high, indicating significant prediction error. However, this is more accurate than when uncleaned data was used.

Polynomial regression using significant terms

I remade my model using significant polynomial terms:

```
>poly(rem, 5)1
>poly(spending_score, 5)1
>poly(spending_score, 5)3
>poly(spending_score, 5)4
>poly(spending_score, 5)5
```

```
> anova(poly_model, significant_poly_model)
Analysis of Variance Table

Model 1: loyalty_points ~ poly(rem, 5) + poly(spending_score, 5)
Model 2: loyalty_points ~ poly(rem, 5)[, 1] + poly(spending_score, 5)[,
    1] + poly(spending_score, 5)[, 3] + poly(spending_score,
    5)[, 4] + poly(spending_score, 5)[, 5]
  Res.Df       RSS Df Sum of Sq      F  Pr(>F)
1   1989 532265306
2   1994 536252029 -5  -3986723 2.9796 0.01101 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- However, in comparison to the original polynomial regression model, a significant degradation in model fit occurred.

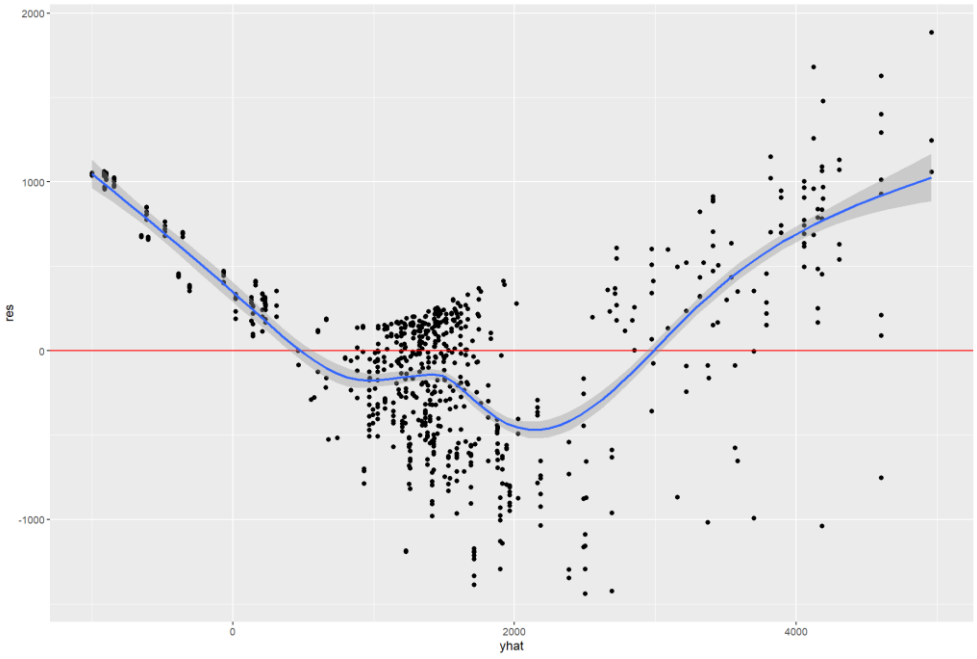
```
> anova(poly_model, model)
Analysis of Variance Table

Response: loyalty_points
                        Df     Sum Sq   Mean Sq F value    Pr(>F)
poly(rem, 5)             5 1251347044 250269409  935.22 < 2.2e-16 ***
poly(spending_score, 5)  5 1508149226 301629845 1127.15 < 2.2e-16 ***
Residuals             1989  532265306    267604
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- The polynomial regression model produced a significant improvement in fit compared to the original MLR model.

Independence of Residuals

A persistent problem persisted still in the polynomial regression model.



- A large discrepancy between residuals and actual values still occurs – suggesting that the residuals are not independent.

Improving the MLR model

I removed any outlier data points to improve the accuracy of the model and satisfy the main assumptions.

I removed loyalty point values >5000 to maintain model accuracy, while also capturing patterns.

I removed remuneration values >100 (outliers).

I fitted this model:

```
Call:
lm(formula = loyalty_points ~ rem + spending_score, data = Subset2)

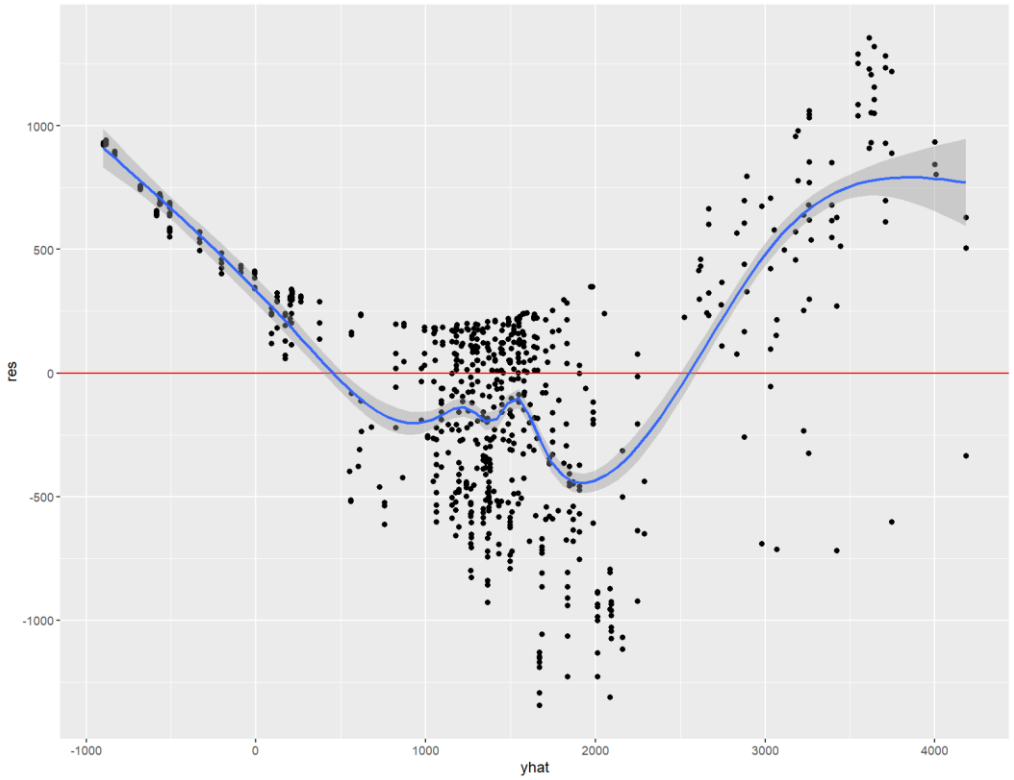
Residuals:
     Min       1Q   Median       3Q      Max
-1343.88  -365.13    51.32   275.40  1356.16

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1476.4657    34.7302  -42.51   <2e-16 ***
rem               31.3285     0.5244   59.74   <2e-16 ***
spending_score    30.3055     0.4288   70.67   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

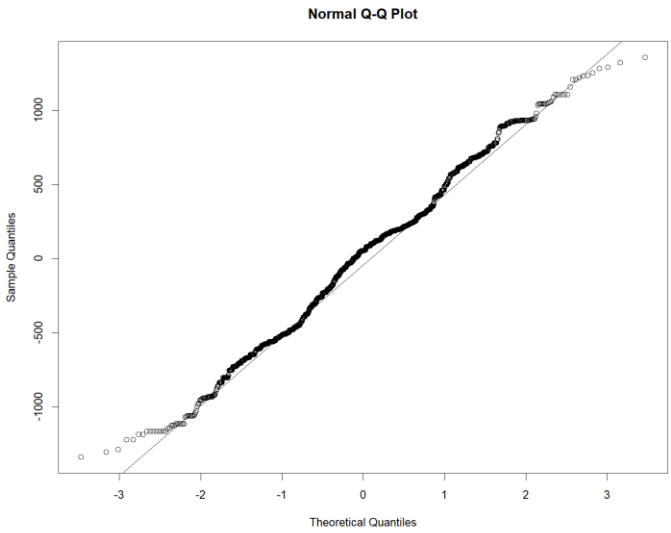
Residual standard error: 481.9 on 1916 degrees of freedom
Multiple R-squared:  0.8086,    Adjusted R-squared:  0.8084
F-statistic:  4048 on 2 and 1916 DF,  p-value: < 2.2e-16

>
> model_summ2 <- summary(model2)
> mean(model_summ2$residuals^2)
[1] 231846.8
```

- The model is a strong fit on the data, as indicated by R-Squared, F-Statistic, and *p-value*.
- MSE still indicates significant prediction error, but a greater accuracy than using uncleaned data.



- The residuals are much closer to the fitted values, than in previous models. The model is more accurate in its predictive power.




```
        Shapiro-Wilk normality test

data:  model_residuals2
W = 0.99262, p-value = 3.089e-08
```

- The Q-Q Plot indicates a lower presence of outliers.
- Shapiro-Wilk Test *p-value* indicates that this model has a more normal distribution of residuals than the original MLR model, however, the distribution is still abnormal.

Removing residual outliers

I remove residual outliers to further improve my MLR model.

```
# I will identify residual outliers and remove
# residuals 2 standard deviations outside the norm.
Subset2$residuals <- resid(model2)
SD2 <- 2*sd(resid(model2))
SD2

# 2 standard deviations is 963.2605.

# I refit the model.
Subset2_cleaned <- Subset2 %>%
  filter(abs(residuals) <= SD2)
```

```
Call:
lm(formula = loyalty_points ~ rem + spending_score, data = Subset2_cleaned)

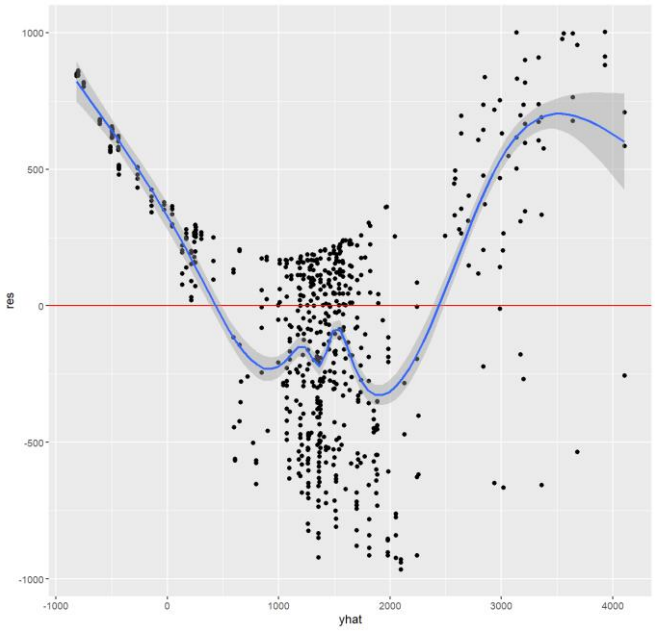
Residuals:
    Min      1Q  Median      3Q     Max
-965.16 -335.72   44.25  255.88 1003.29

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1379.7657    32.2053  -42.84   <2e-16 ***
rem               30.5768     0.5008   61.05   <2e-16 ***
spending_score    29.1357     0.4075   71.50   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

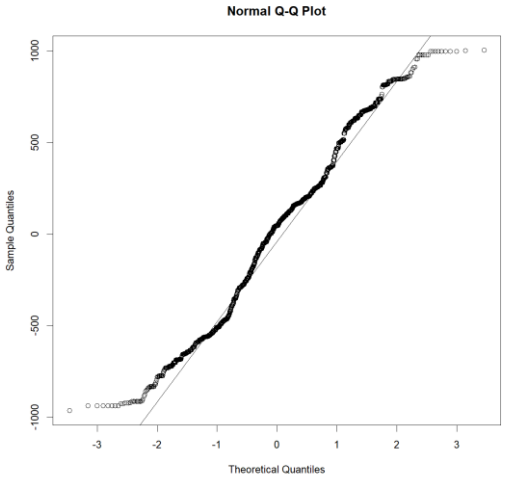
Residual standard error: 434.8 on 1842 degrees of freedom
Multiple R-squared:  0.8224,    Adjusted R-squared:  0.8223
F-statistic:  4266 on 2 and 1842 DF,  p-value: < 2.2e-16

>
> model_summ2cleaned <- summary(model2_cleaned)
> mean(model_summ2cleaned$residuals^2)
[1] 188774.1
```

- R-Squared, F-Statistic, *p-value* indicate a strong fit.
- MSE is much lower than all previous models, however is still a significant prediction error.



- Residuals are much closer to fitted values.



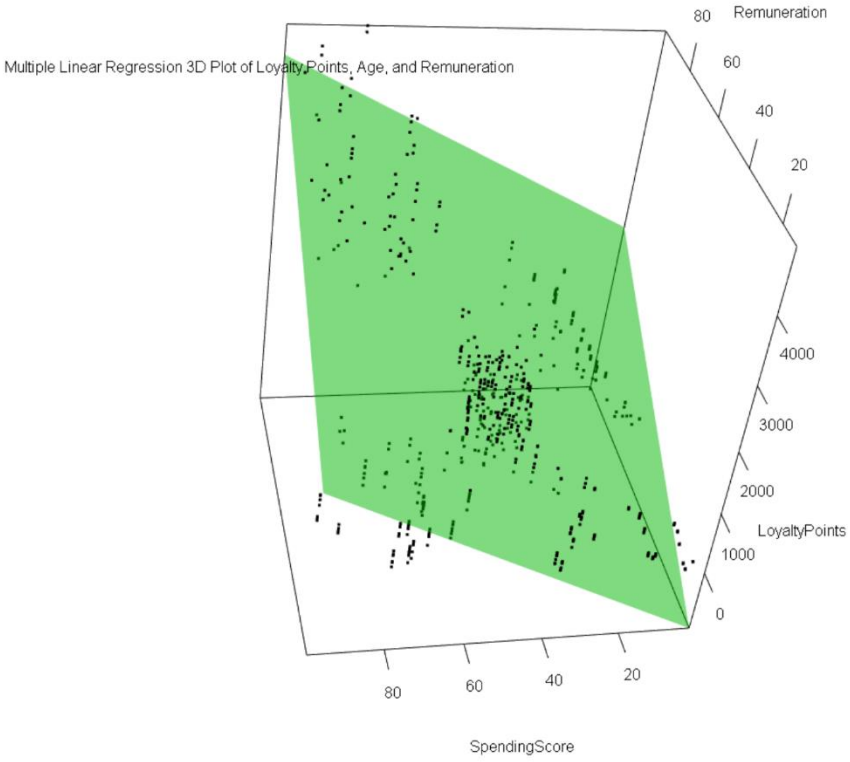

```
        Shapiro-Wilk normality test

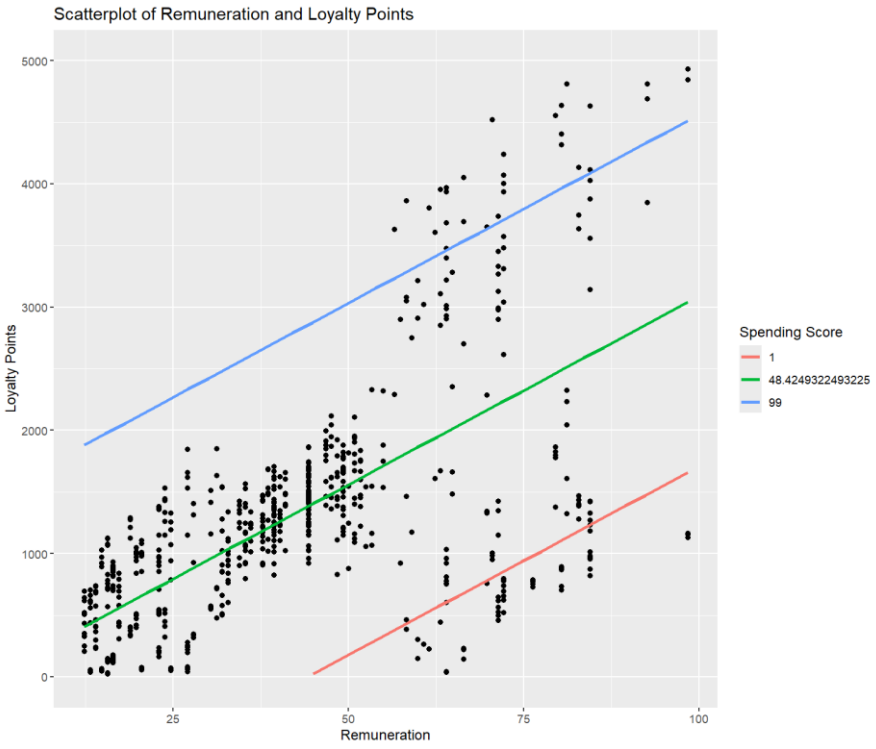
data:  model_residuals3
W = 0.98354, p-value = 9.868e-14
```

- However, the distribution of residuals is much more abnormal than with previous models.

While the normality of distribution is still greater than previously, since the model has a greater predictive accuracy, in the context of the investigation, I shall proceed with this model.

Visualizations





- We can see that as remuneration and spending score are both linked to greater loyalty point accumulation.
- Two distinct sub-populations exist:
    - High remuneration, and low spending score, and loyalty points.
    - High spending score, but low remuneration, and loyalty points.

Summary

- An MLR model was produced that captures much of the underlying patterns. The compromise of this however was the presence of abnormal residual distribution, indicating that the model is not entirely accurate.
- The patterns confirm that remuneration, spending score, and loyalty points are positively correlated with each other.

Q5

I created a binary logistic regression model to predict if customers are high earners (>2000 loyalty points).

$$let\ y = \begin{cases} 1 & \text{if } y > 2000 \\ 0 & \text{if } y \leq 2000 \end{cases}$$

$$P(y=1|x) = \frac{1}{1+e^{-(\beta 0+\beta 1x1+\beta 2x2+\beta 3x3+\beta 4x4)}}$$

$$P(loyaltypoints=1|x) = \frac{1}{1+e^{-(\beta_0 \cdot remuneration+\beta_2 \cdot spendingscore+\beta_3 \cdot gender+\beta_4 \cdot education)}}$$

I made an initial model:

```
Call:
glm(formula = loyalty_points ~ ., family = "binomial", data = train.data)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)    -35.96171    8.64519  -4.160 3.19e-05 ***
rem              0.26817    0.02901   9.245  < 2e-16 ***
spending_score   0.27422    0.03087   8.884  < 2e-16 ***
education1       1.28608    7.71906   0.167  0.86768
education2       2.38986    7.70003   0.310  0.75628
education3       1.30881    7.70811   0.170  0.86517
education4       1.32043    7.69249   0.172  0.86371
gender1          2.06320    0.60041   3.436  0.00059 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1610.02  on 1600  degrees of freedom
Residual deviance:  174.38  on 1593  degrees of freedom
AIC: 190.38

Number of Fisher Scoring iterations: 10

>
```

```
Call:
glm(formula = loyalty_points ~ ., family = "binomial", data = train.data)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)    -32.61243    3.69444  -8.827  < 2e-16 ***
rem              0.26817    0.02901   9.245  < 2e-16 ***
spending_score   0.27422    0.03087   8.884  < 2e-16 ***
education0      -1.28608    7.71906  -0.167  0.86768
education2       1.10378    0.92424   1.194  0.23237
education3       0.02274    0.97173   0.023  0.98133
education4       0.03435    0.89538   0.038  0.96940
gender0         -2.06320    0.60041  -3.436  0.00059 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1610.02  on 1600  degrees of freedom
Residual deviance:  174.38  on 1593  degrees of freedom
AIC: 190.38

Number of Fisher Scoring iterations: 10
```

- From this, I found that remuneration, spending score, and gender are significant in predicting my outcome variable.

I split my dataset into training and test sets. I then predicted the probability of y=1 for each observation in the test set.

```
> probabilities_model <- training_model %>% predict(test.data, type = "response")
> head(probabilities_model)
            3            14            15            20            28            30
8.005356e-12 1.637526e-03 1.018718e-11 4.083359e-01 1.288519e-05 1.250188e-01
```

I rounded each predicted value to either 1 or 0.

Accuracy testing

```
Confusion Matrix and Statistics

          Reference
Prediction   0   1
         0 316   3
         1   3  77

               Accuracy : 0.985
                 95% CI : (0.9676, 0.9945)
    No Information Rate : 0.7995
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.9531

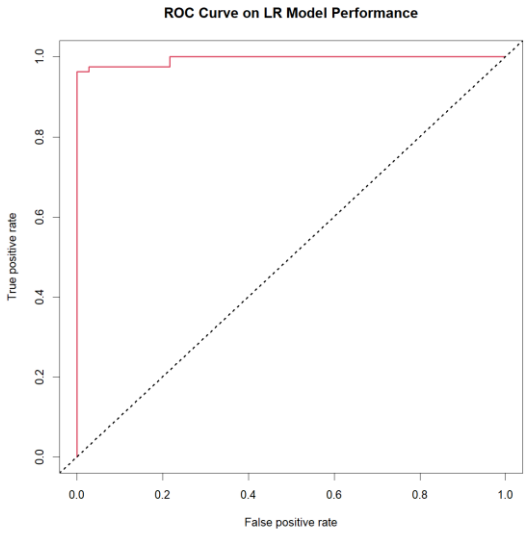
 Mcnemar's Test P-Value : 1

            Sensitivity : 0.9906
            Specificity : 0.9625
         Pos Pred Value : 0.9906
         Neg Pred Value : 0.9625
             Prevalence : 0.7995
         Detection Rate : 0.7920
   Detection Prevalence : 0.7995
      Balanced Accuracy : 0.9765

       'Positive' Class : 0

>
```

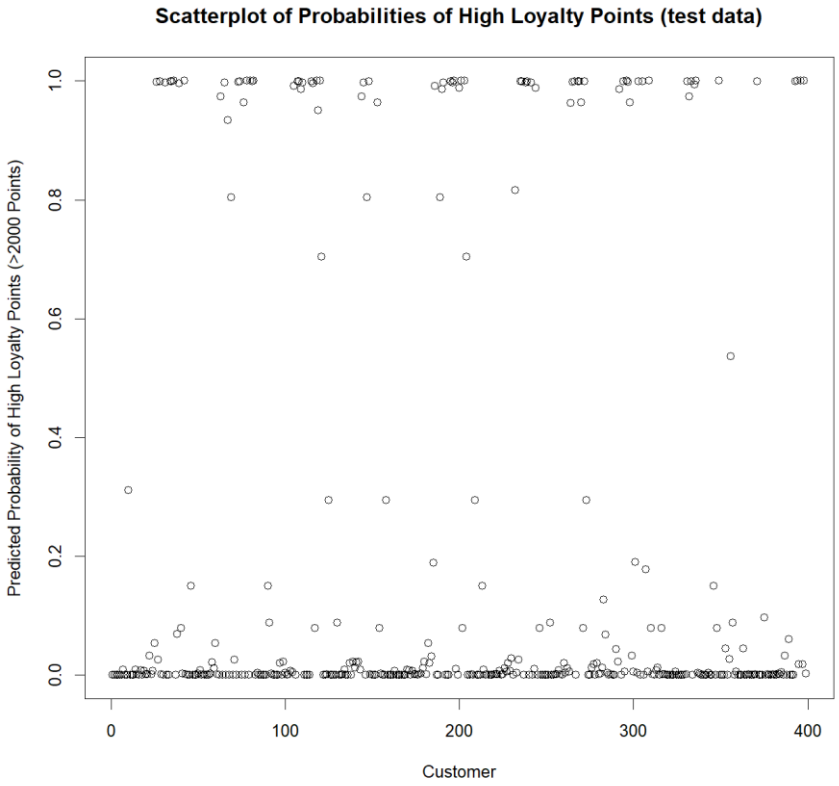
- The model is highly accurate, predicting true positives and true negatives 98.5% correctly.



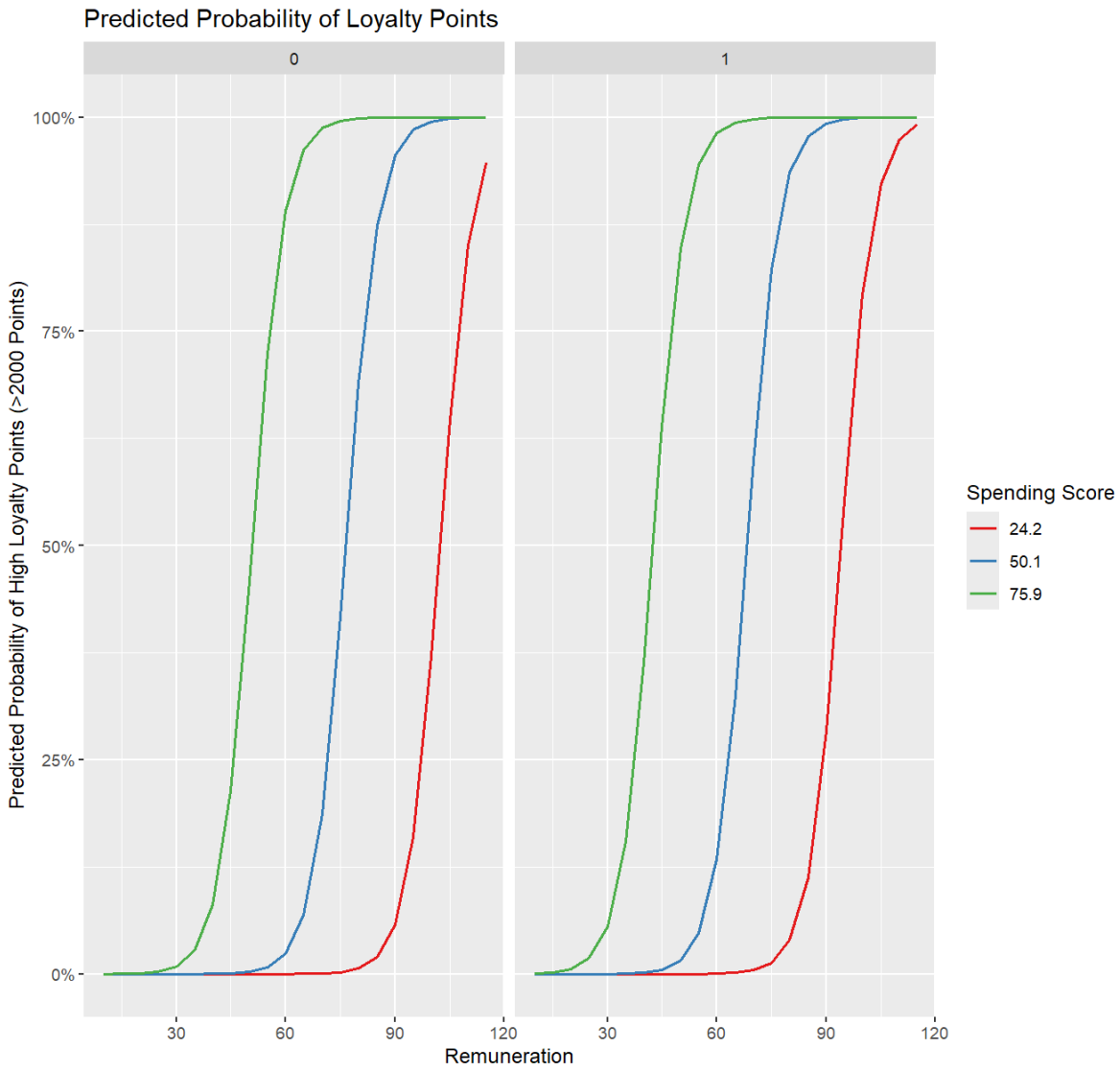

```
[1] "AUC:  0.994239811912226"
```

- ROC Curve and AUC value indicate near perfect predictive accuracy.

Visualizations

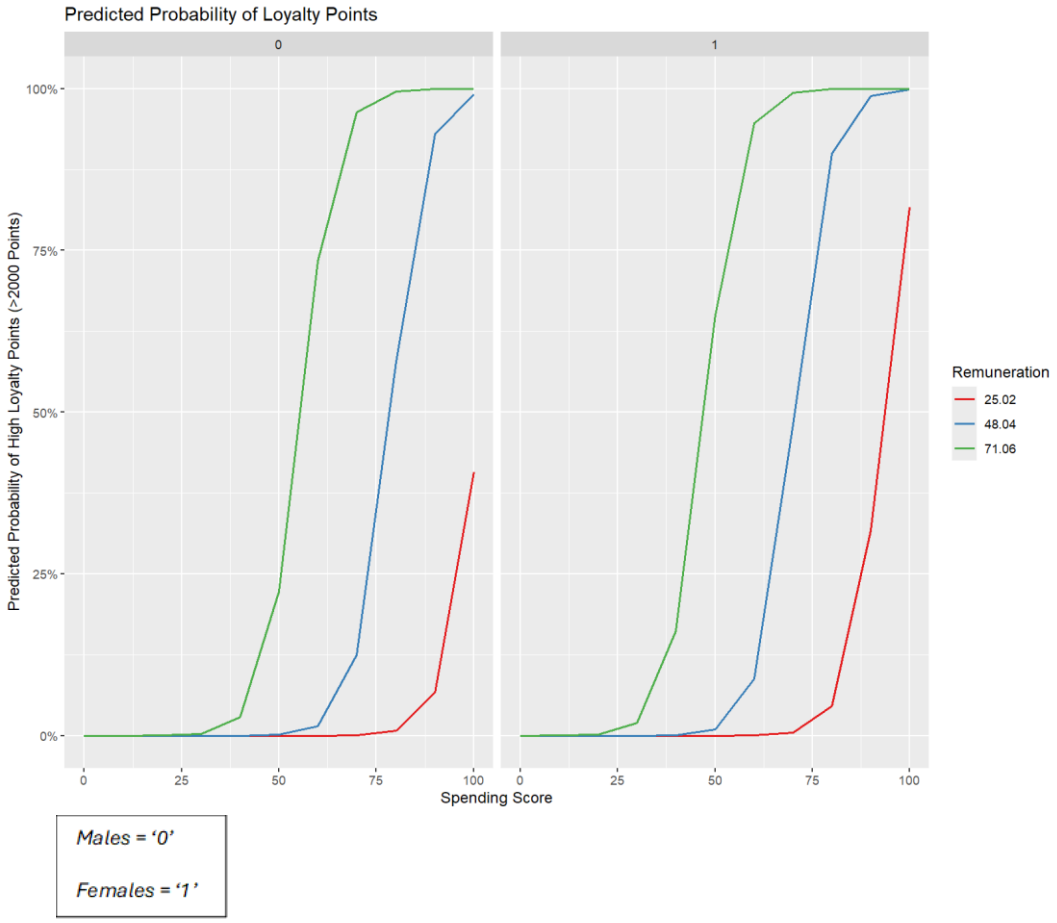


- The model predicts most customers to be low loyalty point customers.
- This is corroborated by the actual raw data, which supports this.



Males = '0'

Females = '1'



- For both males and females, greater remuneration is associated with a higher probability of being a high point customer. This probability is compounded by a higher spending score.
- However, when comparing males and females with similar spending scores, females are more likely to be high point customers. Males with high spending score and low remuneration are unlikely to be high point customers.
    - This group of males who behave in this trend may be those who comprise this cluster of high spending score, but low remuneration, and loyalty point values.

Summary

- Logistic regression model proved to be highly accurate.
- Granular insights were derived into customer trends by remuneration, spending score, and gender, in relation to loyalty points.

Q6

I explore the overall sentiment of the reviews and products left by the customers.

I explored the data within two variables – 'reviews' and 'summary'.

<u>Top 15 words</u>

I found the top 15 most common words in both sets.

I normalised my data, by removing stopwords, and lemmatizing words, in order to produce only the top 15 most common meaningful words.

Word	Frequency
game	1689
great	595
fun	553
one	532
play	502
like	414
love	332
get	319
really	318
cards	301
would	300
tiles	298
good	293
time	291
book	274

'reviews'

Word	Frequency
stars	466
five	381
game	318
great	293
fun	217
love	93
good	92
four	58
like	54
expansion	52
kids	50
cute	45
book	43
one	38
awesome	36

'products'

- We see generally that the most frequent words in both sets are very positive. Customers generally enjoy the product.
- 'game', however, is the most frequent word, which is quite neutral, however most other words are positive.

However, it would provide more insight to look into more granularity. Word frequency does not paint a full picture. For example, one of the most popular words in both sets is ‘one’ – this is ambiguous (‘One Star’, or ‘One of my favourite games’?)

Review sentiment

I analysed the reviews using VADER Sentiment Analysis. Because VADER analyses sentiment by also considering caps and punctuation, I did not alter these. However, many of reviews were hidden behind HTML links, which I pre-processed, applying a user-defined function.

```
def clean_html(html):
    soup = BeautifulSoup(html, "html.parser")
    for data in soup(['style', 'script', 'code', 'a']):
        data.decompose()
    return ' '.join(soup.stripped_strings)
```

```
vader['review'] = vader['review'].apply(lambda x: clean_html(x))
```

For each review, I found the compound polarity score (ranging from -1 (extremely negative) to +1 (extremely positive)).

20 Most Positive Reviews

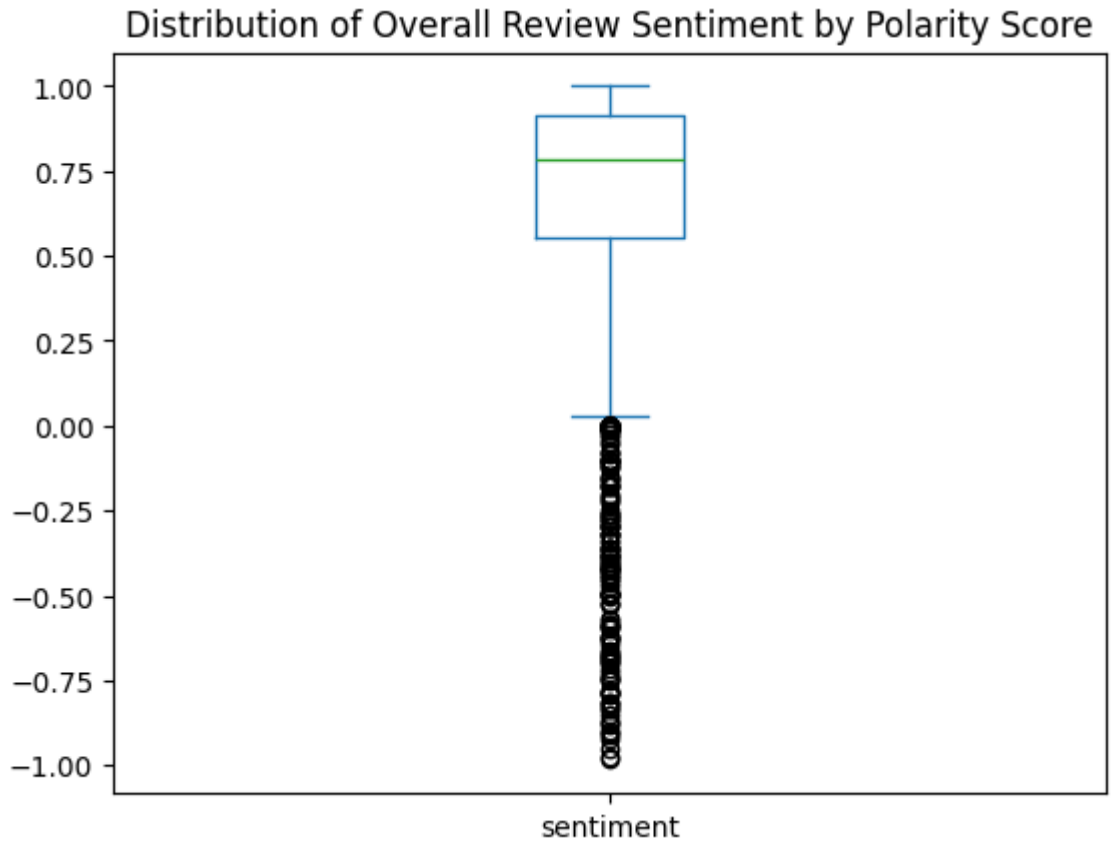
review	sentiment
Disclaimer: Bought this from a local store. Pa...	0.9996
Whenever I see this game on my shelf, I get a ...	0.9994
This kit is AWESOME! My 5-year old daughter an...	0.9992
If you only employ one creativity-enhancing re...	0.9992
As a dad of two boys Im always on the lookout ...	0.9991
I grew up playing Monopoly. Lots of people di...	0.9988
Lords of Waterdeep was awesome, and Scoundrels...	0.9985
Lords of Waterdeep: Scoundrels of Skullport is...	0.9984
If you are a fan of Dungeons and Dragons or ot...	0.9979
We own this game as well as Castle Ravenloft. ...	0.9975
The short, short, version: Wrath of Ashardalon...	0.9974
Wrath of Ashardalon gets everything right for ...	0.9972
Doctor Who: The Card Game was created by Marti...	0.9970
I would recommend this game to any fan of fant...	0.9966
Who am I:\nMiddle aged married guy who loves S...	0.9966
Monopoly is fun, but it takes FOREVER to play,...	0.9966
We were SOOO excited to see the EASTER Story E...	0.9965
I just bought this game with my 8 year old son...	0.9963
Acquire may not be as old as Monopoly, but it ...	0.9962
I was skeptical about 'Castle Ravenloft' despi...	0.9962

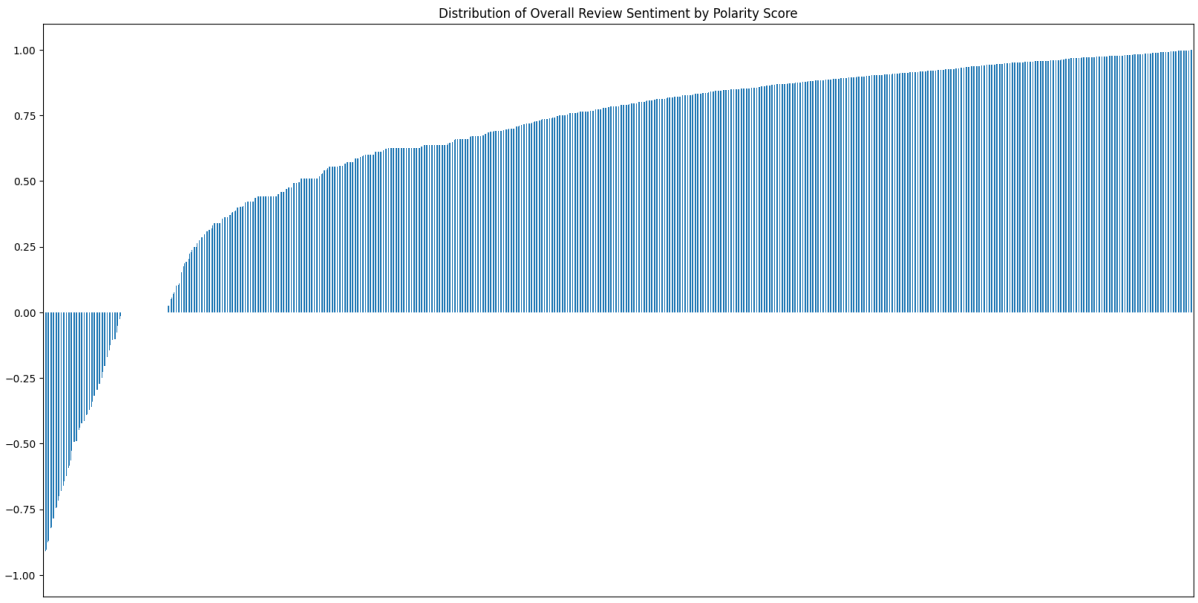
20 Most Negative Reviews

review	sentiment
The One Ring is a very innovative RPG set in M...	-0.9836
Here is my review, cross-posted from boardgame...	-0.9747
I bought this thinking it would be really fun ...	-0.9520
This is a fun game for kids. It's basically th...	-0.9238
Its really UNO type game but anger control stu...	-0.9106
A crappy cardboard ghost of the original. Har...	-0.9052
The Game is amazing.\nThe new version is not.\...	-0.8984
The Ball of Whacks can be a valuable tool for ...	-0.8806
My 8 year-old granddaughter and I were very fr...	-0.8762
firstly, this game is EXCELLENT! the previous ...	-0.8689
I thought I was getting a new product, but the...	-0.8518
This is horrible! The directions are very hard...	-0.8347
Nice art, nice printing. Why two panels are f...	-0.8334
It's really annoying because when you click th...	-0.8211
I found that this card game does the opposite ...	-0.8179
It's Uno with questions about anger. It's an o...	-0.8126
Not a hard game to learn but not easy to win.	-0.7946
If I could give this egg zero stars I would. I...	-0.7845
For those just getting started in the DnD worl...	-0.7841
The product description specifies that this pr...	-0.7838

- It is tedious to examine each review. It may be inaccurate for a person to analyse each review in these subsets to find the commonalities between positive and negative reviews.
- The ‘summary’ set may be easier to derive sentiment patterns.

I visualized the distribution of overall review sentiment.





- A more obvious pattern emerges here: Most reviews are highly positive – however, a significant contingent is very negative.

Summary sentiment

In analysing the 'summary' sentiments, I encountered a problem. Many of the rows contained summaries such as 'One Star' or 'Five Star', however, VADER calculated the compound polarity of such summaries as 0.0.

```
The sentiment value of the sentence :"Five Stars" is : 0.0
```

To solve this problem, I added appropriate weighting to certain strings, then updated the SentimentIntensityAnalyzer lexicon.

```
new_words = {
    'Five Stars': 1.0,
    'Four Stars': 0.8,
    'Three Stars': 0.0,
    'Two Stars': -0.8,
    'One Star': -1.0
}

sia.lexicon.update(new_words)
```

```
# Returning the defined compound score for summaries which
# are now updated from the lexicon.
def get_score(sentence):
    score = sia.polarity_scores(sentence)['compound']
    for word in new_words:
        if word in sentence:
            score = new_words[word]
            break
    return score

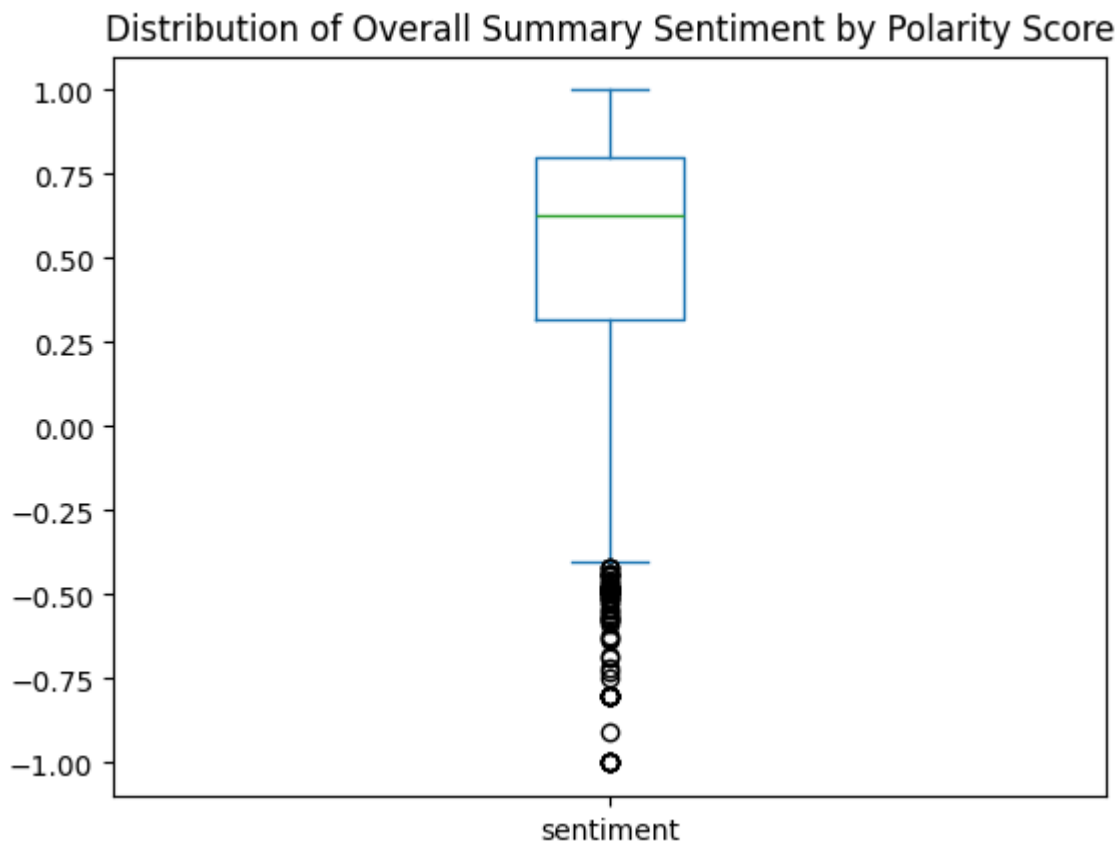
vader_summary['sentiment'] = vader_summary['summary'].apply(get_score)
```

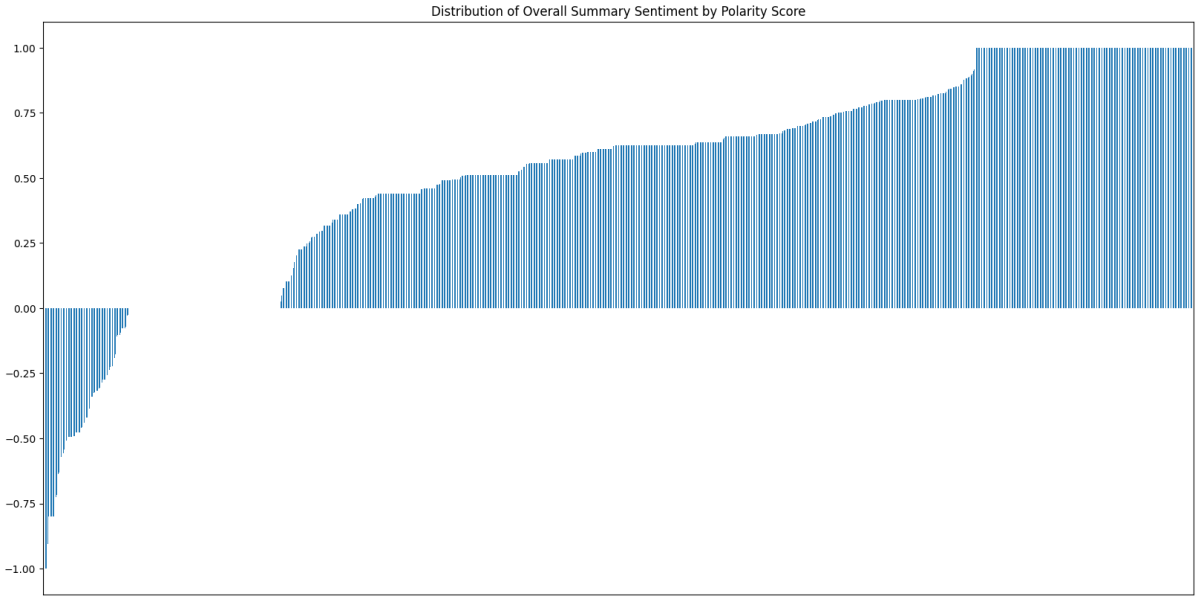
However, this resulted in a case where the top 20 positive and negative summaries were ‘Five Stars’ and ‘One Star’ respectively.

Top 20 Positive Summaries	sentiment
Wow! What a great set of tiles for such a great price. A great starter set.	0.9545
Great for playing, great for creative workshops too!	0.9215
Wrath of Ashardalon: great investment for an avid RPG fan who enjoys tabletop as well as CRPGs	0.9186
Easy to learn, great fun to play!	0.9184
Easy, FUN!, fast and thoroughly enjoyable for adults as well as kids age eight and above.	0.9103
... loves stickers and she loves Peppa so this was perfect..	0.9074
Great quality, very cute and perfect for my toddler	0.9073
but the kids really like it and helps start good discussion	0.8988
Fun, friendly, beautiful game!	0.8932
Great game, great value.	0.891
Great game. Value for the price is great also	0.891
I wish all those who buy better luck. Fairly easy to understand and plenty of ...	0.8885
He loves construction vehicles so I am sure he will love this also	0.8876
Excellent stickers. My grand daughter loves Peppa Pig	0.886
Fantastic Set, great flexibility. Read Review for tip!	0.8858
A useful and fun expansion to an already awesome game	0.8834
Great birthday gift, very cute	0.8832
Hours of good fun here- very well made and stays interesting for ...	0.8832
Fun fun fun!	0.8805
Purchased as a gift for my great nephew who is fascinated by equipment/trucks/tractors	0.8779

Top 20 Negative Summaries	sentiment
A crappy cardboard ghost of the original. Hard to believe they did this, but they did. Shame on Hasbro. Disgusting.	-0.9052
The TARDIS, The Doctor, River Song, Amy, Rory Fight Every Enemy - In Cards!	-0.75
No 20 sided die	-0.7269
Defective- poor QC	-0.7184
Box totally destroyed!!!!!!!!!!!!!!!!!!!!!!!!	-0.6869
Then you will find this board game to be dumb and boring	-0.6808
before this I hated running any RPG campaign dealing with towns because it ...	-0.6369
but it gets repetitive and the students start to get bored after about half a round	-0.631
Worst quality adult board game I've even seen	-0.6249
Horrible! Nothing more to say Would give zero stars ...	-0.5848
Anger Control game	-0.5719
Anger Control Game	-0.5719
Adds a (needed) amount of tension in a tense fantasy world.	-0.5719
Really small disappointed!	-0.5676
Great game, but lazy and boring board design.	-0.5647
It's UNO for the angry!	-0.5562
Too bad, this is not what I was expecting.	-0.5423
Bad Quality-All made of paper	-0.5423
Very cute, but poor quality stickers	-0.5266
Not the best quality.	-0.5216

- The positive summaries were high in polarity, indicating that the game is fun, of great value, intuitive for children, and engaging.
- The negative summaries were not as extreme in their negative polarity as compared to the positive summaries. The negative summaries highlighted mostly frustration arising from the game, and quality issues with the equipment.





- The trend in summary polarity score distribution mirrored that of reviews.
- However, a larger contingent were entirely neutral, perhaps due me updating the compound polarity of ‘Three Stars’ to 0.0.

Summary

- The ‘summary’ set provided a clearer insight into customer sentiments.
- The product is generally positive among the customers, however, significant number report issues concerning equipment quality and frustration.